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MULTI-PROTOCOL LABEL SWITCHING

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Abstract

MPLS (Multi Protocol Label Switching) specified mechanisms to manage traffic flows of different granularities, such as flows between different hardware, machines, or even flow between different applications, independent of layer 2 and 3 protocols, and offers a means to map IP addresses to simple,fixed-length labels used for packet forwarding using different packet switching technologies.One way to protect the MPLS infrastructure is to separate the resources used by MPLS services from resources used for other purposes. In an MPLS network, it is possible to separate the resources used for the control plane Of those used for the data plan. This means that its resources can be protected and physically isolated from any other equipment to protect user data while the control plane uses network resources that can be accessed by operators to configure the network.

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1 Unit 1 : Introduction

1.1 History of MPLS

As the internet grew in popularity, organizations looked for an efficient way to perform packet forwarding. Bandwidth demands increased, but label-switching mechanisms struggled to handle the load. Traditional methods, such as IP switching and tag switching, require each router to independently determine a packet's next hop by inspecting its destination IP address before consulting its routing table. This slow process involves hardware ressources and introduces the potential of degraded performance for real-time applications, such as voice and video. Traditional routers needed to scale more effectively to meet the bandwidth needs of the modern internet and avoid slow speeds, jitter and packet loss.

In 1997, the Internet Engineering Task Force (IETF) Multiprotocol Label Switching working group formed to create standards to help fix the issues around internet traffic routing. MPLS was developed as an alternative to multilayer switching and IP over asynchronous transfer mode (ATM). MPLS routers don't look up routes in routing tables, which helps boost the speed of network traffic. As MPLS techniques were developed and adopted throughout the early 2000s, the protocol became widely adopted.

MPLS can work in a multiprotocol environment, such as ATM, frame relay, Synchronous Optical Network and Ethernet. MPLS continues to evolve as backbone network technologies evolve, and the IETF working group still works on MPLS protocols and mechanisms. MPLS also played a significant role in the support of legacy network technologies, as well as newer technology based on IP networks.

1.2 Definition of MPLS

MPLS is a data transport mechanism based on label switching, where labels are inserted at the ingress of the MPLS network and removed at its egress. Basically, this insertion occurs between the data link layer and the network layer in order to transport protocols such as IP .[1]

MPLS labels are advertised between routers so that they can build a label-to-label mapping. These labels are attached to IP packets, allowing routers to forward traffic by looking at the label and not the destination IP address. Packets are forwarded by label switching rather than IP switching.[2]

That is why MPLS is referred to as a layer "2.5" protocol as shown in Figure 1. This protocol has further evolved to provide a unified data transport service for customers using a packet switching technique. MPLS can be used to transport virtually any type of traffic, such as voice or IPv4, IPv6 packets and even Ethernet or ATM frames.[1]

MPLS enables large-scale deployment to carry different types of traffic while respecting the associated operating constraints and on a single infrastructure illustrated in Figure 1.

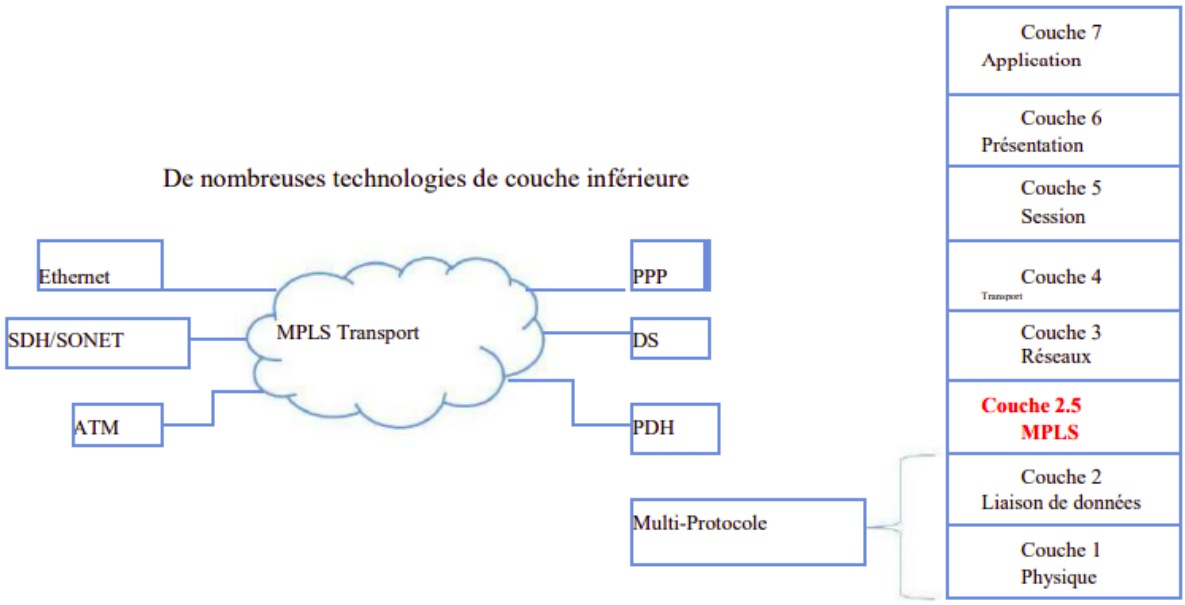


FIGURE 1 – POSITIONING MPLS IN THE OSI MODEL

1.3 Needs and Justifications for MPLS

In the mid-1990s, the volume of data transmitted over the Internet increased sharply with the rapid development of IP networks. Due to hardware limitations at the time, IP technologies, using the longest match rule, had to use software for route lookups, which resulted in poor forwarding performance. This also limited the development of IP networks. In this context, the IETF proposed the MPLS protocol to improve the forwarding performance of routing devices on IP networks.[3]

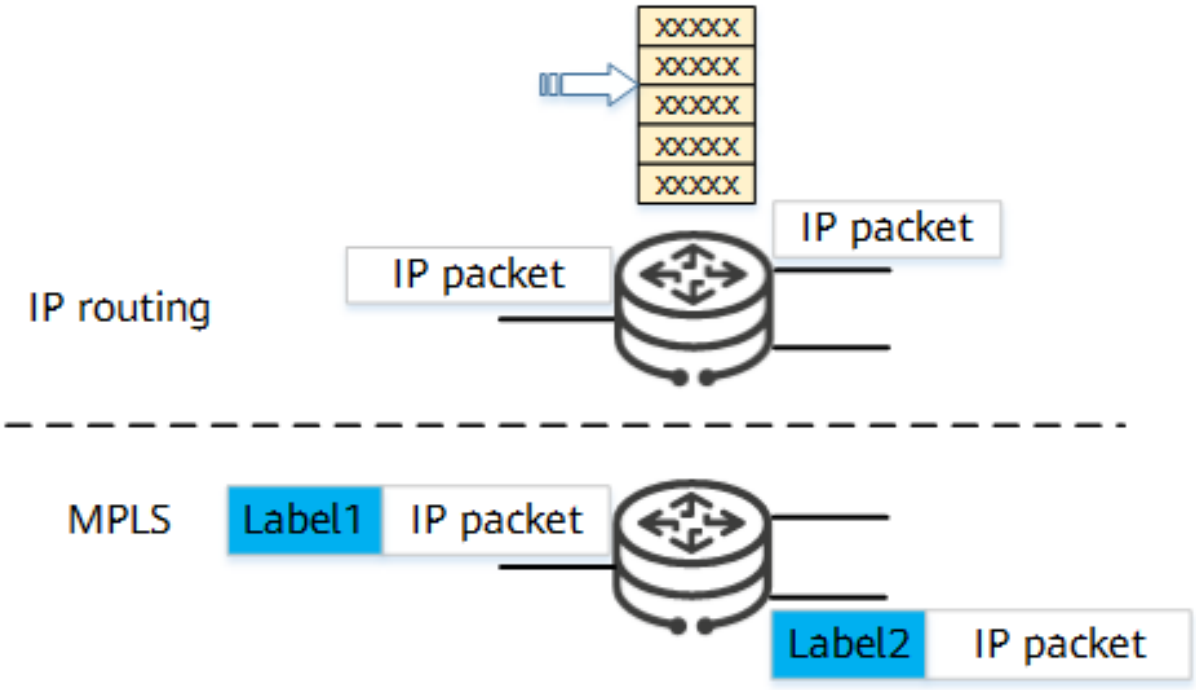


FIGURE 2 – Comparison between IP routing and MPLS forwarding

1.3.1 Limitations of IP Routing

In traditional computer networks, routers face a major challenge when handling data packets. These packets contain only destination IP addresses. No other information, about their nature, or the best way to route them, is added. This limitation is mainly due to the BGP protocol. Thus, each router relies only on the destination IP address to route packets. This operation leads to a significant inefficiency in traffic management. Each router must have information about the destination prefix, and the paths taken are not always optimized.[4]

1.3.2 Addressing the Limitations of IP Routing with MPLS

MPLS presents a novel idea—"labels"—to get around this restriction. These labels serve as an extra layer that defines the data packets' class of service. To include more information, they are allocated to every packet. Following that, these are routed to routers in the same class. The destination prefix information is only required to be known by the originating and terminating routers. As a result, consistent traffic management is ensured since packets of the same class always take the same route. The MPLS network can now handle packets based on their unique properties thanks to this method. These could include the application kind, their place of origin, or the necessary quality of service (QoS).[4]
The protocol can therefore be broken down into 4 stages :

— **Packet Labeling :** When a data packet enters an MPLS network, the ingress router assigns it a label.

— **Packet Forwarding :** MPLS routers examine the labels of incoming packets to determine the appropriate path to follow. Routers use these labels to make fast and efficient routing decisions.

— **Label Stripping :** When a packet reaches the egress router of the MPLS network, the router removes the label from the packet header.

— **Public Network Forwarding :** The packet is then forwarded to the final network using traditional IP routing information .

1.4 Advantages and Benefits of MPLS

MPLS offers many benefits to network administrators and businesses. Here are the main ones [5] :

— **Reliability :** Label-based routing in the private network ensures that packets are reliably delivered to their destination. In addition, MPLS allows for the prioritization of traffic for different packet types, such as routing real-time video packets through low-latency paths. This reliability is guaranteed through a service-level agreement (SLA), which also ensures that the MPLS provider fixes the outage or pays a penalty.

— **High performance :**MPLS's dedicated infrastructure ensures high quality, low latency, and low jitter performance. This ensures efficiency and a good user experience. It is also essential for real-time communications such as voice, video, and important information.

— **Prioritizing flows with QoS :** MPLS offers excellent Quality of Service (QoS) due to its ability to prioritize data flows. This allows bandwidth to be allocated efficiently as needed. This quality is therefore beneficial for real-time applications such as VoIP or email.

— **Simplification of Architecture :** Unlike some more complex technologies like SD-WAN, MPLS is based on a simple architecture with a single link layer. This configuration provides an optimized solution for secure switching between locations. This is one of the differences between MPLS and SD-WAN.

— **Multi-Protocol :** One of the great benefits of MPLS is its technology independence. It is designed to work with a variety of underlying protocols. This versatility meets the diverse requirements of modern networks.

— **Security :**This protocol isolates traffic to provide high security. Attacks are more difficult because packets are sent over predetermined routes. A key element of digital risk management is a VPN.

In conclusion, MPLS has revolutionized network flow management by introducing labels to optimize traffic, improve quality of service, and simplify network management. By offering a unified and versatile solution, it continues to be an essential technology for companies concerned with improving performance in their network communications.

2 MPLS Components :

Understanding the basic ideas that underpin MPLS's operation is crucial to comprehending what it is. These ideas form the foundation of this technology and describe how it gets beyond the drawbacks of conventional.Key concepts of MPLS :

2.1 MPLS label :

A label is a four-byte (32-bit) identifier that conveys the predetermined forwarding path of the packet in an MPLS network. While a network address specifies an endpoint, a label specifies the paths between endpoints. This latter capability allows MPLS to decide the optimal path for a given packet. Labels can also contain information about the QoS and priority level of a packet. An MPLS label has 4 bytes. The following figure shows how it is encapsulated :[6][5]

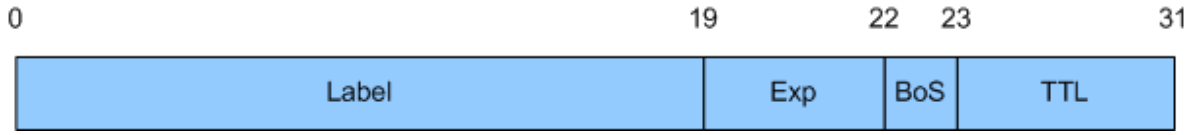


FIGURE 3 – MPLS Label Encapsulation Structure

An MPLS label consists of the following four fields :

— **Label value :** A 20-bit field that identifies a label value.

— **Exp :** A 3-bit field used for extensions.These are used for QoS, normally the IP precedence value of the IP packet will be copied here.

— **Bottom of Stack (S) :** A 1-bit field that identifies the bottom of a label stack. if this flag is set, it signifies that the current label is the last in the stack.

— **Time to Live (TTL) :** An 8-bit field, just like in the IP header, this is the time to live field. You can use this for traces in the MPLS network. Each hop decrements the TTL by one.

Between the network layer and the link layer is an MPLS label. Where an MPLS label is encapsulated in a packet is depicted in the following figure. All link layer protocols support MPLS labels.



FIGURE 4 – MPLS Label Encapsulation Position

Another name for an MPLS label stack is an MPLS multilayer label. As can be seen in the following figure, it includes an ordered set of MPLS labels. "Top of the stack" or outer label is the label that is located close to the Layer 2 header ; "bottom of the stack" or inner label is the label that is located close to the IP header. Labels are processed from last to first out, starting at the top of the label stack. [3]

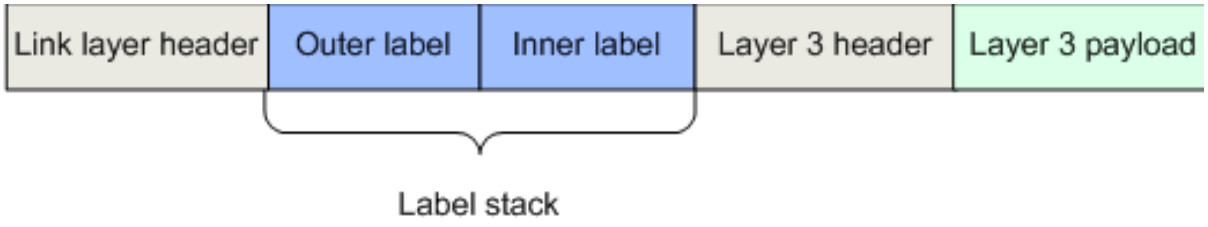


FIGURE 5 – MPLS Label Stack

2.2 Label Switching Router (LSR) :

Label switch routers (LSRs) or transit routers are MPLS routers that route based solely on the label. In the center of an MPLS network is this kind of router. It is in charge of changing the labels that packets are routed using.

LSR contains two types of routers :[8]

— **PE Router (Provider Edge) :** The PE router serves as a point of entry and departure for client traffic and is situated at the edge of the MPLS network. It converts IP packets into MPLS packets by adding MPLS headers to incoming packets and connecting to the customer networks. In order to forward outgoing packets to clients, it is also in charge of deleting MPLS headers. For customer isolation, the PE router is essential.implementing customer-specific QoS policies.

— **P Router (Provider) :** The MPLS network's central component is the P-Router, also known as the label switching router. It is in charge of switching MPLS packets according to MPLS labels after receiving them from PE routers or other P-Routers. Using only MPLS labels enables faster and more effective switching because the P-Router does not read the packets' IP addresses. It guarantees the network's packet forwarding is optimized rather than actively participating in routing decisions.

When an LSR receives a packet, it uses the label included in the packet header as an index to determine the next hop on the label-switched path (LSP) and a corresponding label for the packet from a Label Information Base. The old label is then removed from the header and replaced with the new label before the packet is routed forward.

2.3 Label edge router (LER) :

A label edge router, also called Provider Edge Router, is an LSR that is located on the edge of an MPLS network and is connected to another network, an access node to the MPLS network. A LER can have multiple ports to access several distinct networks, each of which can have its own switching technique.[7] LERs play an important role in the implementation of labels. We distinguish two types of LER :

- **Igress-LER (Ingress Label Edge Router) :** This is a router located at the entrance to the MPLS network and in which we will assign labelsfor the flow arriving from the customer's network.
- **Egress-LER (Egress Label Edge Router) :** It does the opposite of Ingress-LER by removing labels from packets coming from the core of the MPLS network.

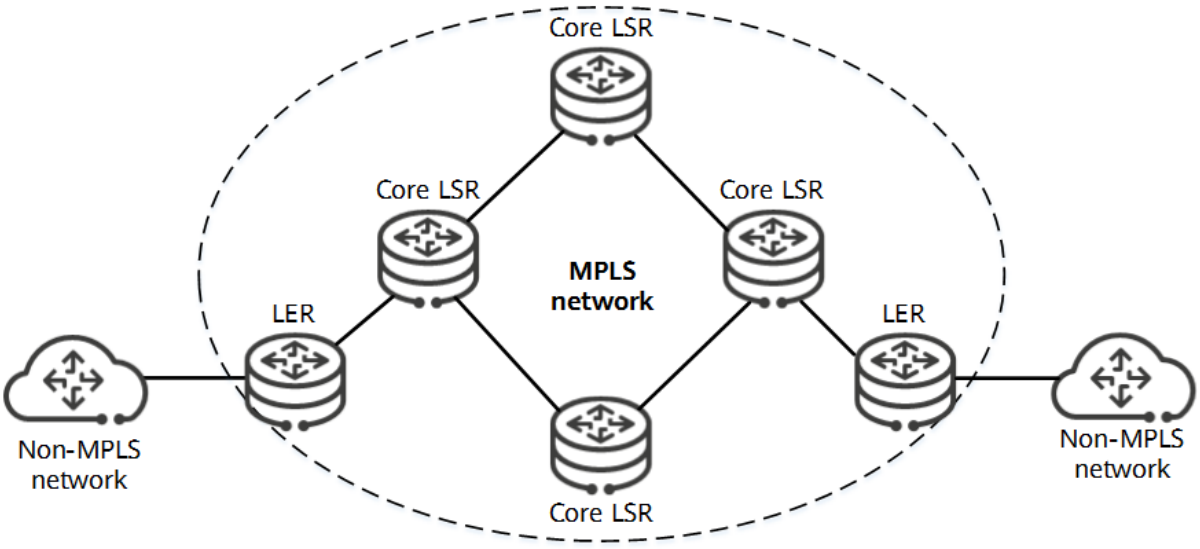


FIGURE 6 – NODES PARTICIPATING IN MPLS : LSR AND LER

2.4 Router CE (Customer Edge) :

On the consumer side, usually at the intersection of the customer and provider networks, is the CE router. It is in charge of using the PE router to connect the client's devices and equipment to the MPLS network. Before packets are sent to the PE router for MPLS encapsulation, the CE router can be set up to implement particular routing policies or perform other traffic management tasks.[8]

These different types of routers in an MPLS network work together to enable fast and efficient packet switching, providing service providers with the ability to deliver robust, quality connectivity services to their customers.

2.5 Forwarding equivalence class (FEC) :

In MPLS, routing is done through equivalence classes. A class represents a flow or set of flows that have the same properties, including the same prefix in the IP address.
All frames of a FEC are treated the same in the nodes of the MPLS network. They are introduced into a FEC at the ingress node and can no longer be distinguished within the class of other flows.[9]

2.6 The Label-Switched Path (LSP) :

An MPLS domain is determined by a set of MPLS nodes on which FECs are determined.

From an MPLS network's ingress to its egress, an LSP is a unidirectional path. Two nearby LSRs on an LSP are referred to as the upstream LSR and downstream LSR, respectively, in the packet forwarding direction. LSR A is the downstream LSR of LSR B, and LSR B is the upstream LSR of LSR A in Figure 7.

LSPs are determined on a domain before data arrival in the most classic case. Two options are used for this purpose [9] :

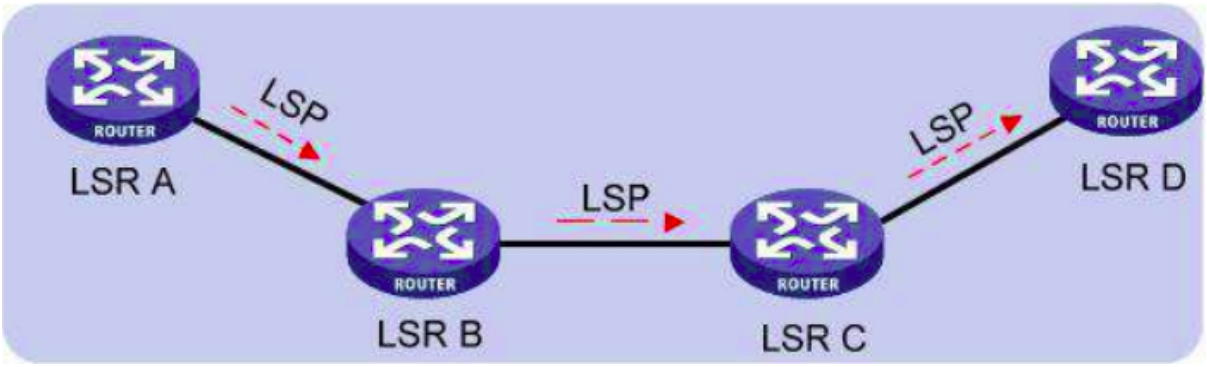


FIGURE 7 – DIAGRAM FOR LSP

(Hop-by-hop) routing : In this case, LSRs select the next hops independently of each other. The LSR uses a routing protocol such as OSPF or for ATM-type subnets for this purpose.

Explicit routing (same as source routing) : In this case, the MPLS domain ingress LER specifies the list of nodes through which the signaling has been routed. The choice of this path may have been constrained by quality of service requirements. The path followed by frames in one direction of communication may be different in the other direction.

3 Forwarding in MPLS : Swap, Push et Pop :

When an unlabeled packet enters the ingress router and needs to be passed on to an MPLS tunnel, the router first determines the forwarding equivalence class (FEC) for the packet and then inserts one or more labels in the packet's newly created MPLS header. The packet is then passed on to the next hop router for this tunnel. When a labeled packet is received by an MPLS router, the topmost label is examined. Based on the contents of the label a swap, push or pop operation is performed on the packet's label stack. Routers can have prebuilt lookup tables that tell them which kind of operation to do based on the topmost label of the incoming packet so they can process the packet very quickly.

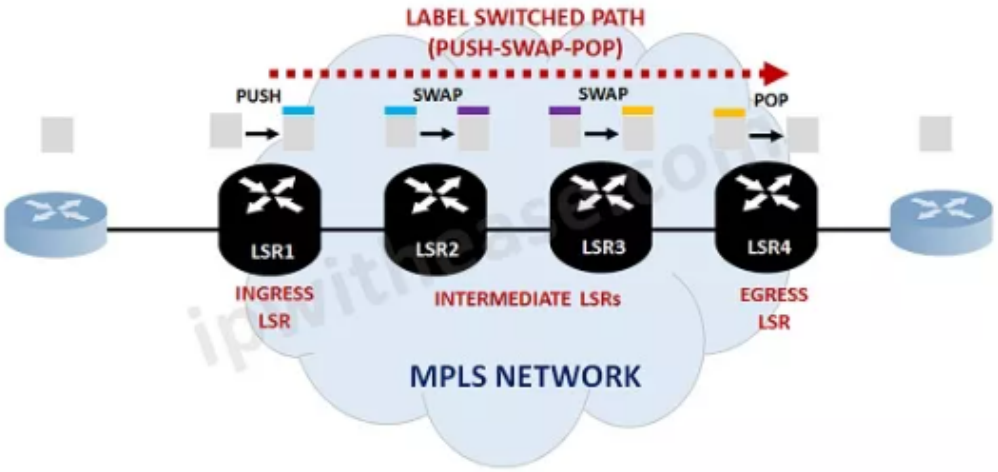


FIGURE 8 – SWITCHING PROCESSING IN CORE MPLS

The basic operations on MPLS labels are as follows[8] :

1. *POP (Pop the Label) :* When an MPLS packet reaches the last LSR of a Label-Switched Path, it performs the POP operation. This means that it removes the MPLS label from the packet header before forwarding it to the final delivery network. POP is essential because it restores the packet to its original IP format, ensuring correct delivery to the final destination.
2. *PUSH (Push the Label) :* When an LSR adds an MPLS label to an incoming IP packet, it performs the PUSH operation. The Provider Edge (PE) router is typically responsible for this task when it receives packets from a client and encapsulates them in MPLS format by adding an appropriate MPLS label to the packet header. This creates the start of the LSP and routes the packet into the MPLS network.
3. *SWAP (Swap the Label) :* When packets are routed along the LSP, LSRs perform the SWAP operation to exchange the current MPLS label with a new label. The choice of the new label is based on the MPLS routing table, which associates a specific label with a destination route. The SWAP operation allows LSRs to efficiently redirect packets along pre-established paths in the MPLS network, providing fast and optimized switching.

4 MPLS Architecture :

The MPLS architecture is divided into two distinct components : the transmission component, also called the "data plane" and the control component, also called the "control plane"[10].

4.1 Control Plane :

Based on the routing protocol, the control plane creates a routing table (Routing Information Base, or RBI). The control plane can use a variety of routing protocols, including OSPF, IGRP, EIGRP, IS-IS, RIP, and BGP. The control plane creates and maintains labels internally and exchanges them with other MPLS nodes via a label exchange protocol. LDP, BGP (used by MPLS VPN), and the older Cisco TDP protocol are examples of label exchange protocols. MPLS uses RSVP to carry out the label exchange.

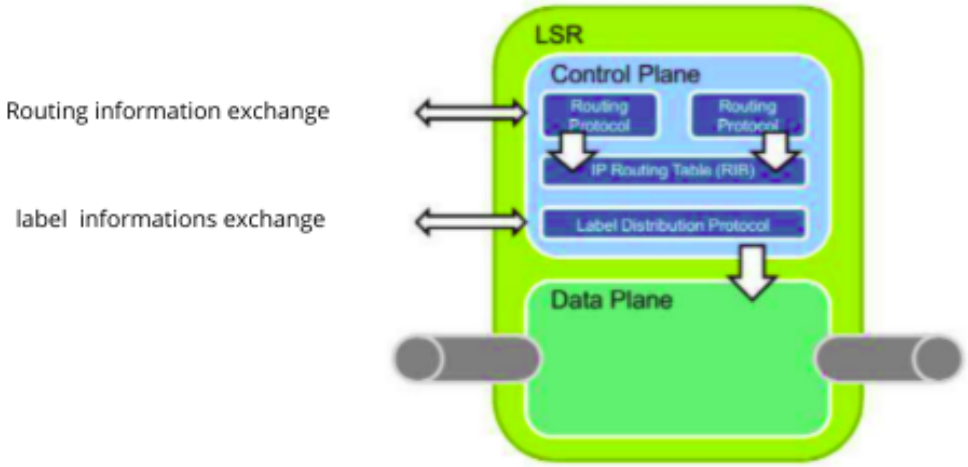


Figure 9 – MPLS ARCHITECTURE : CONTROL PLANE

Additionally, the control plane depends on two tables : a Label Forwarding Information Base (LFIB) based on the label and RIB exchange protocol and a Forwarding Information Base (FIB) based on the data in the RIB. Each network prefix's label values and affiliations with the outgoing interface are listed in the LFIB table.

4.2 Data Plane :

The data plane takes care of shipping based on destination addresses or labels.
The data plane is also known as the routing plane. It is a simple routing engine that is independent of the type of routing protocol or label exchange protocol used.
The data plane forwards packets to the appropriate interface based on the information in the FIB and LFIB tables.

4.3 Forwarding Structures :

The data plane on a router is responsible for forwarding packets based on decisions made by routing protocols, which extend into the router control plane. The data plane on an MPLS router consists of two forwarding structures :

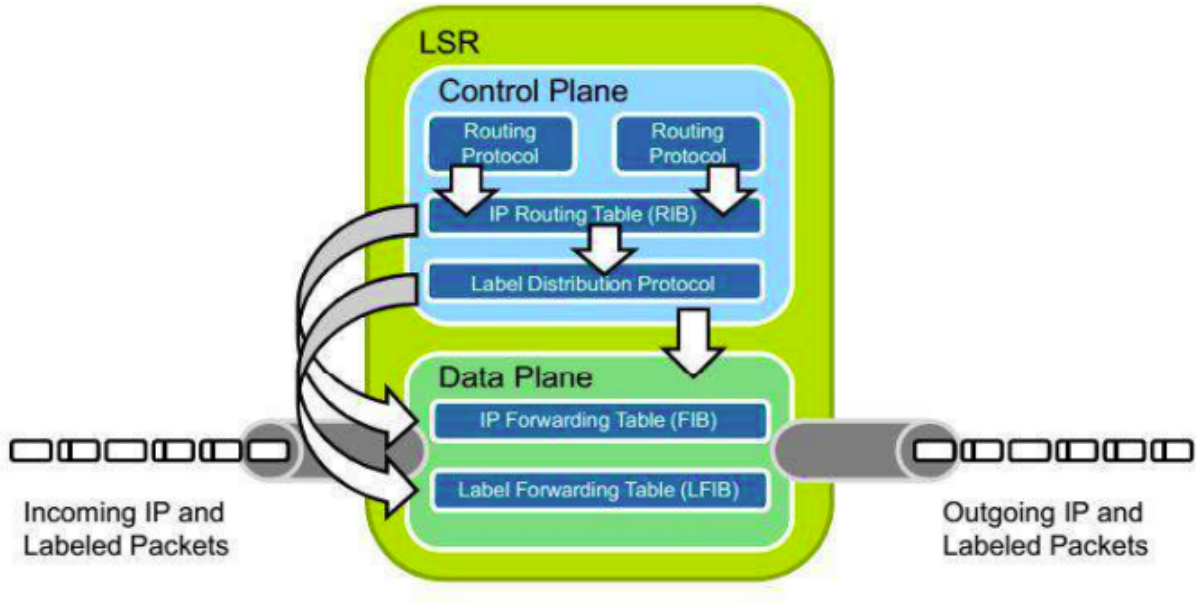


FIGURE 10 – MPLS ARCHITECTURE : DATA PLANE

4.3.1 Forwarding Information Base -FIB- :

When a router is turned on, IP packets are forwarded via the FIB in accordance with the routing protocols' judgments. A routing table comprising destination networks, next hops, exit interfaces, and pointers to Layer 2 addresses is loaded into the FIB. If an exit interface is enabled for MPLS, the FIB on an MPLS-enabled router additionally has an outgoing label. When an IP packet is received, the router performs a FIB lookup. Depending on the outcome, it may either deliver another IP packet or apply a label.

4.3.2 Label Forwarding Information Base -LFIB-

The LFIB is used when a labeled packet is received. It contains the incoming and outgoing label, the outgoing interface, and the next hop information. When an LFIB lookup is done, the result can be to swap a label and send a labeled packet or to remove a label and send an IP packet.

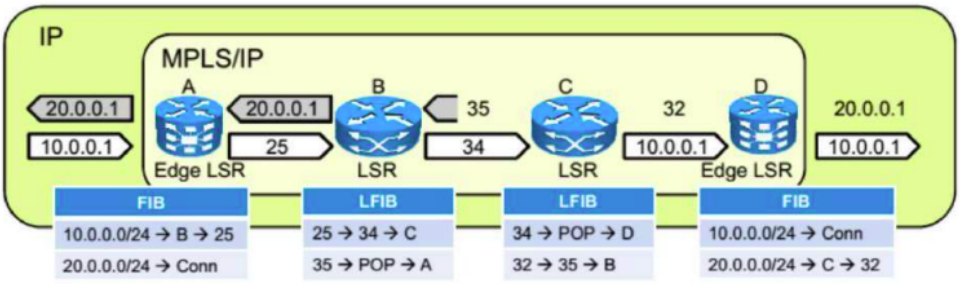


FIGURE 11 – TRANSFER STRUCTURE IN AN IP/MPL CORE

These combinations of packet transmission are possible :

-A received IP packet (FIB) is forwarded based on the destination IP address and it is sent as an IP packet.
-A received IP packet (FIB) is forwarded based on the destination IP address and it is sent as a tagged packet.
-A received labeled packet (LFIB) is forwarded based on the label, the label is exchanged, and the labeled packet is sent.
-A received labeled packet (LFIB) is forwarded based on the label, the label is removed and the IP packet is sent.

4.4 Example of MPLS Architecture :

The figure 12, illustrates an example of protocols used in the MPLS control plane and LFIB in the data plane. In the LSR example architecture, the control plane uses these protocols : An OSPF routing protocol that receives and sends information on the IP network 10.0.0.0/8.
A LDP label exchange protocol that receives the label 24 used for packets with the destination address 10.0.0.0/8. A local label 17 is generated and sent to upstream neighbors so that these neighbors can label the packets with the appropriate label.

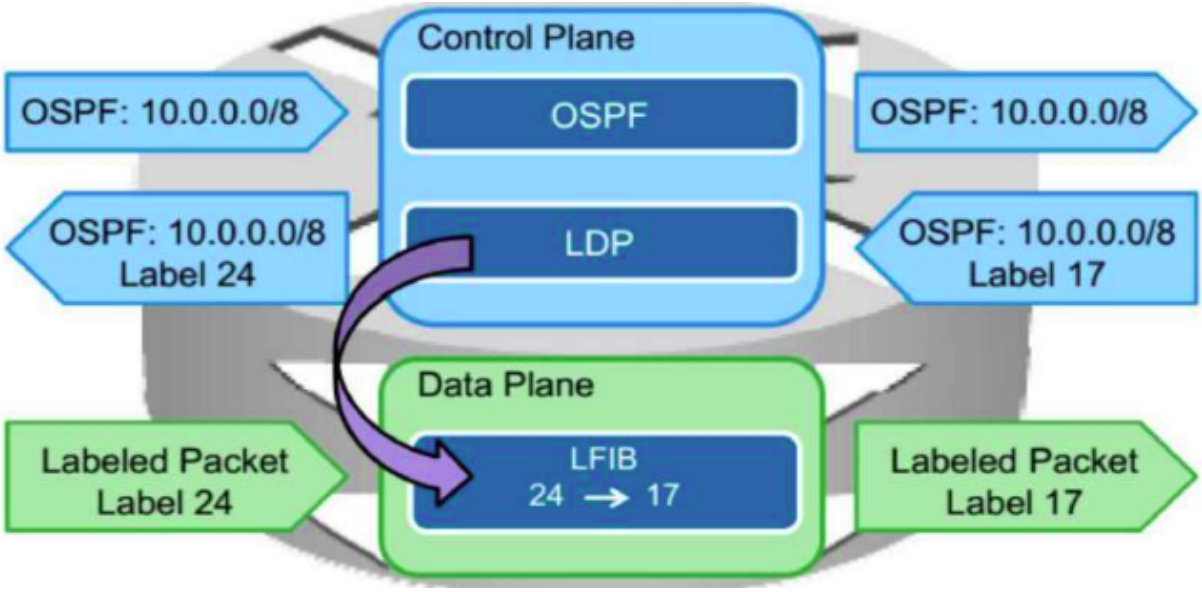


FIGURE 12 – MPLS ARCHITECTURE EXAMPLE

The data plane uses an LFIB to forward packets based on labels as follows : The LFIB receives an entry from LDP, where label 24 is mapped to label 17. When the data plane receives a packet marked with a 24, it replaces label 24 with label 17 and forwards the packet through the appropriate interfaces.

5 How MPLS works :

Data transmission is carried out on paths called LSPs. They are established before the transmission of data (control-driven) or upon detection of a flow (data-driven) that wishes to cross the network. The labels included in the frames are distributed using a

signaling protocol, the most important of these protocols is LDP, but we also use RSVP, possibly associated with a routing protocol, such as BGP or OSPF. The frames carrying the IP packets carry the labels from one node to another.[9]
An input label therefore makes it possible to determine the FEC through which the flow transits. This solution resembles the notion of virtual circuit in ATM, where virtual circuits are multiplexed.here we have a multiplexing of all virtual circuits inside a FEC, so that we can no longer distinguish the virtual circuits.[9]
The LSR examines the label and sends the frame in the indicated direction. We can thus see the crucial role played by the LERs, which assign labels to packet flows that allow the frames to be switched to the correct virtual circuit. The label only has meaning locally, since it is modified on the next link.[9]
Once the packet is classified in a FEC, a label is assigned to the frame that will transport it.This label determines the exit point by chaining the labels, in the case of classic frames, such as LAP-F of Frame Relay or ATM, the label is positioned in the DLCI or in the VPI/VCI.[9]
The signaling required to deposit the value of the labels of the path determined by a FEC can be managed either at each flow or by a control environment independent of the user flows. The latter solution is preferable in the case of large networks, due to its scalability.

5.1 Label distribution :

MPLS standardizes several methods to achieve label distribution. Distribution indicates that each node has its own labels and must match them with the labels of its neighbors.[5] The methods for distributing label references are as follows :
Topology-based : is based on topology, it uses messages intended for routing management like OSPF and BGP.
Request-based : is flow-based, it uses a request to open a path for an IP flow. This is the case of RSVP.
Traffic-based : is traffic-based, upon receipt of a packet, a label is assigned to the frame carrying it.
Topology-based and flow-based methods correspond to one, while traffic-based methods correspond to data.
Routing protocols, including IGP, have been enhanced to carry an additional label. Similarly, the RSVP protocol has a version associated with MPLS that allows it to carry alabel, Where the new version is RSVP-TE, which allows the opening of paths taking into account the network resources
The IETF has also standardized a new signaling protocol, LDP, to manage the distribution of labels. Extensions to this protocol, such as CR-LDP (Constraint-based Routing-LDP) allow the choice of routes followed by FEC clients with a predefined quality of service.
The main signaling protocols are :
LDP, which maps unicast IP addresses to labels.
RSVP-TE and CR-LDP, which open routes with quality of service.
PIM (Protocol Independent Multicast), which maps multicast IP addresses to associated labels.
BGP, which is used to determine labels in virtual private networks.

5.2 Flow aggregation :

Streams from different interfaces can be gathered and switched to a single label, if they go to the same output direction. This corresponds to stream aggregation, this technique is already used in ATM networks, in which a circuit can aggregate several streams from different input nodes to a common point, where the streams are disaggregated.[9]
The purpose of stream aggregation is to reduce the number of labels used or, that is to prevent switching tables from becoming too large.

5.3 Signaling :

As explained earlier, several label distribution mechanisms, called signaling, can be implemented in the nodes of an MPLS network, including the following :
Label request : An LSR issues a label request to its downstream neighbors, which can be linked to the value of a FEC. This mechanism can be used from node to node up to the egress node of the MPLS network.
Label Correspondence : In response to a label request from an upstream node, an LSR sends a referral from an already established matching mechanism to the output node.
The following figure gives an illustration of these two mechanisms.

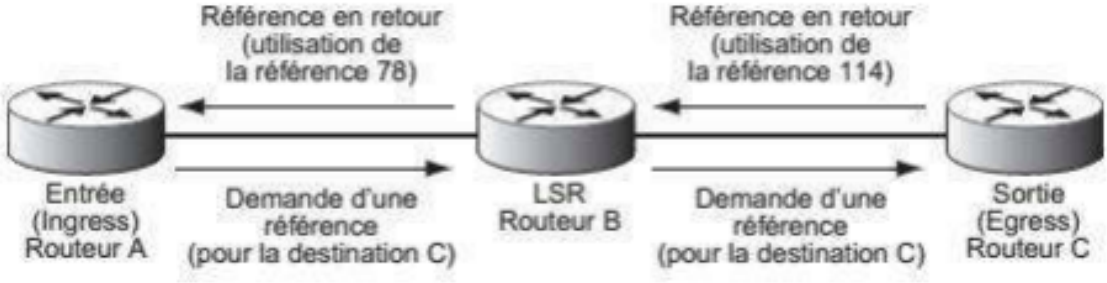


Figure 13 – MPLS SIGNALING MECHANISM

5.4 Label Distribution Protocol (LDP) :

The label distribution protocol that is most frequently utilized in MPLS is called LDP. Unicast and multicast addresses are taken into consideration by this protocol. The output nodes control the explicit routing, and the TCP protocol is used for the exchanges to guarantee a satisfactory level of quality.
Adjacent messages and messages indicating the labels are the two sorts of communications that are approved. When there is agreement among the neighboring nodes, the second sends the label values. The first enables the querying of the nodes that are directly accessible from the origin node. These messages are encoded in the traditional manner that enables the description of an object : we specify the object's type in the first field, its total length in the second, and its value in the third. The TLV (Type Length Value) is the name given to this encoding.
Routing is done, as we have seen, by FEC. The granularity of the labels, that is to say

the size of the flows that use the same label, results from the size of the FEC : if there are few FECs, the flows are large and the granularity is high. And if there are many FECs, the flows are small, and the granularity is fine.
The destination can also be a particular application on a given machine, which gives a high granularity. This last case is illustrated by the figure 13.

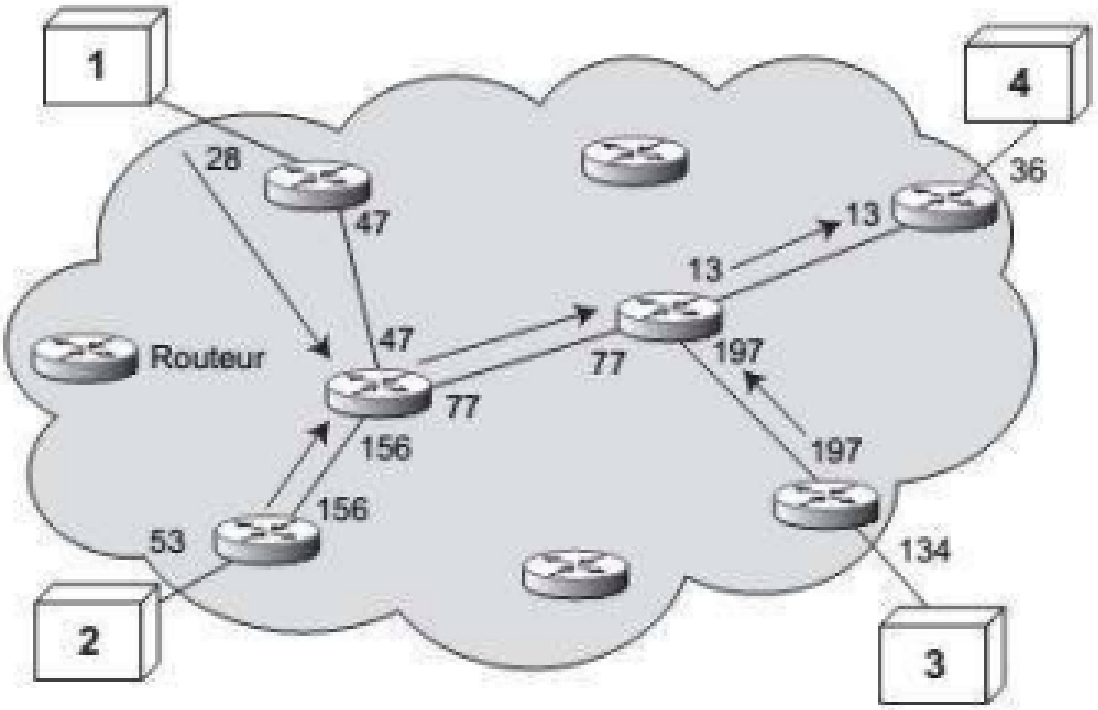


FIGURE 14 – FEC EQUIVALENCE CLASS IN MPLS NETWORKS

6 Unit 2 : Virtual Private Network (VPN)

6.1 What is an MPLS VPN ?

Before defining a Multiprotocol Label Switching Virtual Private Network (MPLS VPN), you need to define a VPN in general.

A VPN is an IP network providing private network services over a public infrastructure ; a set of sites authorized to communicate with each other privately via the Internet or other public or private networks.

MPLS VPN, which stands for Multiprotocol Label Switching Virtual Private Network, is a set of techniques using multiprotocol label switching (MPLS) to establish virtual private networks (VPNs). This approach offers remarkable flexibility for the transmission and routing of diverse network traffic by exploiting the capabilities of an MPLS backbone.

Ultimately, MPLS-based VPNs are created at level 3 and are based on the peer-to-peer model. This model enables the service provider and customer to exchange Layer 3 routing information. The service provider relays data between the customer's sites without the latter's intervention.[12]

6.2 Principal Technologies :

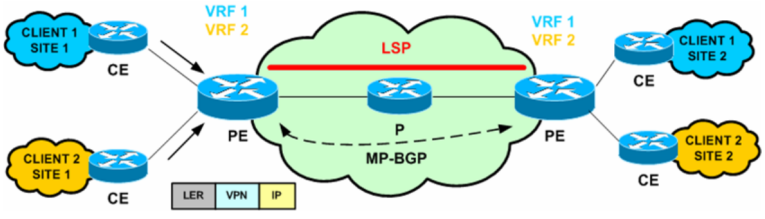
There are four principal technologies that make it possible to build MPLS-based VPNs :

- Multiprotocol Border Gateway Protocol (MP-BGP) between PEs carries CE routing information
- Route filtering based on the VPN route target extended MP-BGP community attribute
- MPLS forwarding carries packets between PEs (across the service provider backbone)
- Each PE has multiple VPN routing and forwarding instances (VRFs)

6.3 How an MPLS Virtual Private Network Works :

Multiprotocol Label Switching virtual private network (MPLS VPN) functionality is enabled at the edge of an MPLS network. The provider edge (PE) device performs the following :

- Exchanges routing updates with the customer edge (CE) device.
- Translates the CE routing information into VPNv4 routes.
- Exchanges VPNv4 routes with other PE devices through the Multiprotocol Border Gateway Protocol (MP-BGP).[12]

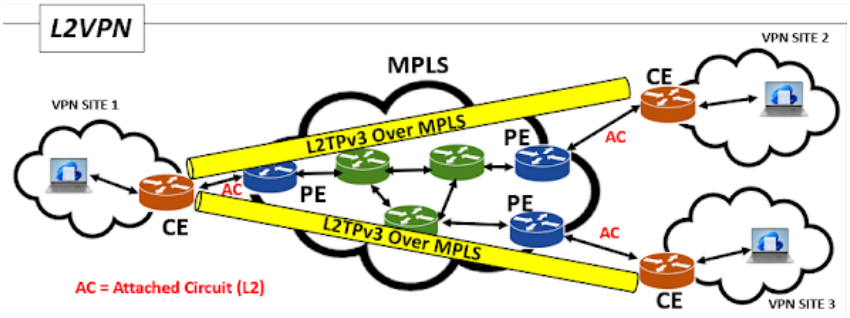


6.3.1 Point-to-Point (Pseudowire) :

Point-to-Point (Pseudowire) : This category of MPLS VPN establishes direct, private connections between two distinct network endpoints. It effectively simulates a dedicated point-to-point link, facilitating secure and efficient communication between remote locations.

6.3.2 Layer 2 (VPLS – Virtual Private LAN Service) :

A Layer 2 MPLS VPN is a term in computer networking. It is a method that Internet service providers use to segregate their network for their customers, to allow them to transmit data over an IP network. This is often sold as a service to businesses.Layer 2 VPNs are a type of Virtual Private Network (VPN) that uses MPLS labels to transport data. The communication occurs between routers that are known as Provider Edge routers (PE routers), as they sit on the edge of the provider's network, next to the customer's network.Internet providers who have an existing Layer 2 network (such as ATM or Frame Relay) may choose to use these VPNs instead of the other common MPLS VPN, Layer 3. There is no one IETF standard for Layer 2 MPLS VPNs. Instead, two methodologies may be used. Both methods use a standard MPLS header to encapsulate data. However, they differ in their signaling protocols.[13]



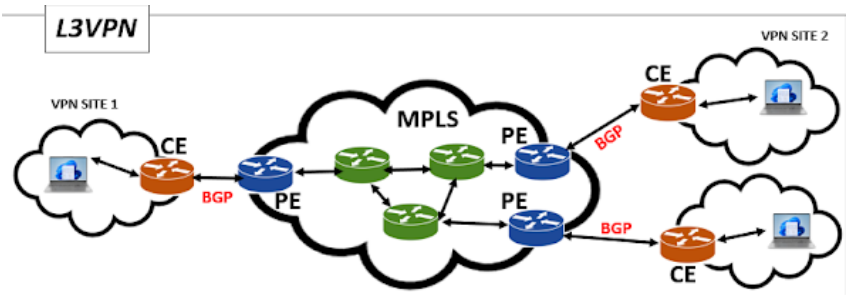
Types of Layer 2 MPLS VPNs :

•BGP-based : The BGP-based type is based on the informational RFC 6624 by Kireeti Kompella from Juniper Networks. It uses the Border Gateway Protocol (BGP) as the mechanism for PE routers to communicate with each other about their customer connections. Each router connects to a central cloud, using BGP. This means that when new customers are added (usually to new routers), the existing routers will communicate with each other, via BGP, and automatically add the new customers to the service.

•LDP-based : The second type is based on the Internet Standard RFC 8077 by Chandan Mishra from Cisco Systems. This method is also known as a Layer 2 circuit. It uses the Label Distribution Protocol (LDP) to communicate between PE routers. In this case, every LDP-speaking router will exchange FECs (forwarding equivalence classes) and establish LSPs with every other LDP-speaking router on the network (or just the other PE router, in the case when LDP is tunnelled over RSVP-TE), which differs from the BGP-based methodology. The LDP-based style of layer 2 VPN defines new TLVs and parameters for LDP to aid in the signaling of the VPNs .

6.3.3 Layer 3 (VPRN – Virtual Private Routed Network) :

Layer 3, or VPRN (virtual private routed network), utilizes layer 3 VRF (VPN/virtual routing and forwarding) to segment routing tables for each customer utilizing the service. The customer peers with the service provider router and the two exchange routes, which are placed into a routing table specific to the customer. Multiprotocol BGP (MP-BGP) is required in the cloud to utilize the service, which increases complexity of design and implementation. L3 VPNs are typically not deployed on utility networks due to their complexity ; however, a L3 VPN could be used to route traffic between corporate or datacenter locations.[14]



6.4 What is the role of Virtual Private network :

MPLS VPNs play a pivotal role in modern networking, empowering businesses to achieve enhanced security, traffic management, and efficiency in their communication infrastructure. The ability to choose from these three distinct MPLS VPN types ensures that organizations can tailor their network solutions to meet their specific operational requirements.

6.5 The different parts of the MPLS VPN are described as follows :

- Provider (P) device-Device in the core of the provider network. P devices run MPLS switching, and do not attach VPN labels to routed packets. The MPLS label in each route is assigned by the provider edge (PE) device. VPN labels are used to direct data packets to the correct egress device.
- PE device-Device that attaches the VPN label to incoming packets based on the interface or subinterface on which they are received. A PE device attaches directly to a customer edge (CE) device.
- Customer (C) device-Device in the ISP or enterprise network.
- CE device-Edge device on the network of the ISP that connects to the PE device on the network. A CE device must interface with a PE device.[12]

6.6 Major Components of an MPLS Virtual Private Network :

An Multiprotocol Label Switching (MPLS)-based virtual private network (VPN) has three major components :

- VPN route target communities-A VPN route target community is a list of all members of a VPN community. VPN route targets need to be configured for each VPN community member.
- Multiprotocol BGP (MP-BGP) peering of VPN community provider edge (PE) devices-MP-BGP propagates virtual routing and forwarding (VRF) reachability information to all members of a VPN community. MP-BGP peering must be configured on all PE devices within a VPN community.
- MPLS forwarding-MPLS transports all traffic between all VPN community members across a VPN service-provider network. A one-to-one relationship does not necessarily exist between customer sites and VPNs. A given site can be a member of multiple VPNs. However, a site can associate with only one VRF. A customer-site VRF contains all the routes available to the site from the VPNs of which it is a member.[12]

6.7 VPN Routing and Forwarding Tables (VRFs) :

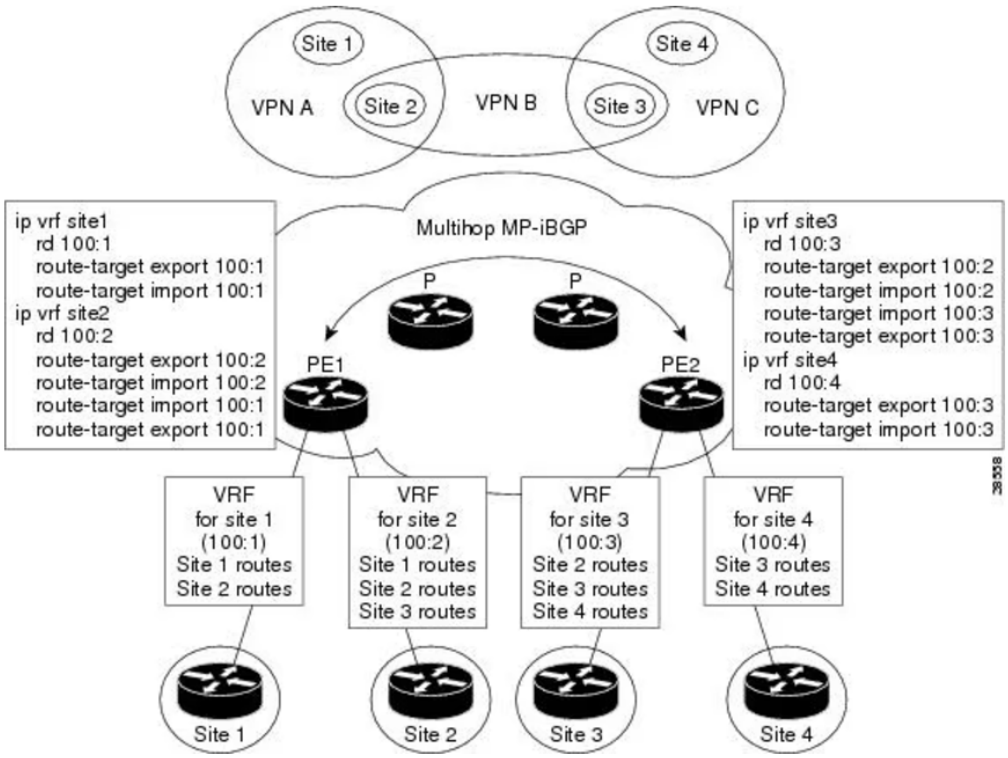
The VPN routing and forwarding table (VRF) is a key element in the MPLS VPN technology. VRFs exist on PEs only. A VRF is a routing table instance, and more than one VRF can exist on a PE. A VPN can contain one or more VRFs on a PE.The VRF contains routes that should be available to a particular set of sites. VRFs use CEF technology, therefore the VPN must be CEF-enabled.

A VRF is associated with the following elements :

- IP routing table
- Derived forwarding table, based on the Cisco Express Forwarding (CEF) technology
- A set of interfaces that use the derived forwarding table
- A set of routing protocols and routing peers that inject information into the VRF

Each PE maintains one or more VRFs. VPNSC software looks up a particular packet's IP destination address in the appropriate VRF only if that packet arrived directly through an interface that is associated with that VRF. The so-called "color" MPLS label tells the destination PE to check the VRF for the appropriate VPN so that it can deliver the packet to the correct CE and finally to the local host machine.[11]

Figure 1-5 VRFs for Sites in Multiple VPNs



Route Distinguisher (RD) and Route Target (RT) are two different concepts used in an IP-MPLS Backbone.

Route Distinguisher (RD) : The RD, local to the PE level, keeps the uniqueness of the prefixes in the common global table of the BGP (each prefix belongs to its VRF, even if there are similarities).

- Format : ASN :nn or IP :nn.
- Example : In the image, for Site 1, the RD is 100 :1.

  Route Target (RT) : The RT is propagated and used to identify the membership of prefixes to their VRFs across the MPLS cloud. It is therefore used to transfer routes between VRFs/VPNs (import and export).
- Example : In the image, Site 1 exports and imports routes with the RT 100 :1.

MP-BGP (Multiprotocol BGP) : MP-BGP (Multiprotocol BGP) is an extension of the BGP protocol. Standard BGP only supports IPv4 unicast address families, whereas MP-BGP supports more than 15 different BGP address families.

RFC4760 defines the extensions for BGP protocol and states that " in order to bring extra functionality to BGP protocol, new BGP AFI (Address Family Identifier) and BGP SAFI (Sub Address Family Identifier) are introduced".

Multiprotocol BGP supports IPv4 and IPv6 address families and their unicast and multicast variants.

MP-BGP also supports Layer 2 VPN address families. EVPN (Ethernet VPN) is a mechanism by which the mac addresses are exchanged between PE devices over the BGP control plane.

All these address families are exchanged between BGP neighbors over a single BGP session in parallel.

BGP neighbors exchange their capability in the BGP open message. If BGP neighborship is set up only for IPv4 BGP address family and later on an extra protocol such as L3 VPN address family is added, the BGP session goes down. That's why careful planning is important to avoid downtime even in maintenance windows since some networks cannot even tolerate planned failure.

If MP-BGP needs to be set up between two BGP nodes (It can be between Route Reflector and Route Reflector Clients in BGP Route Reflector topologies, a new address family needs to be explicitly activated.

• MP-iBGP (Internal BGP) : Used inside the provider's network to share VPN routes between PE routers.

7 Benefits of an MPLS Virtual Private Network :

MPLS VPN is a powerful tool that merges the speed and efficiency of MPLS with the security and privacy of VPNs. It works by creating a virtual tunnel for data to pass securely and quickly through the public network. Although it doesn't inherently provide encryption, it's possible to add an extra layer of security by encrypting the data before it enters the MPLS network. Understanding these concepts and how they work together can help individuals and businesses alike make more informed decisions about their networking needs.

Multiprotocol Label Switching virtual private networks (MPLS VPNs) allow service providers to deploy scalable VPNs and build the foundation to deliver value-added services, such as the following :Connectionless Service,Centralized Service ,Scalability ,Security ,Ease of Creation ,Flexible Addressing ,Integrated QoS Support and Straightforward Migration .

8 Unit 3 : MPLS Traffic Engineering (TE)

MPLS Traffic Engineering (TE) allows network operators to define how traffic flows through an MPLS core network. You can steer traffic to maximize the efficient use of available bandwidth between routers.

8.1 Traffic engineering before MPLS

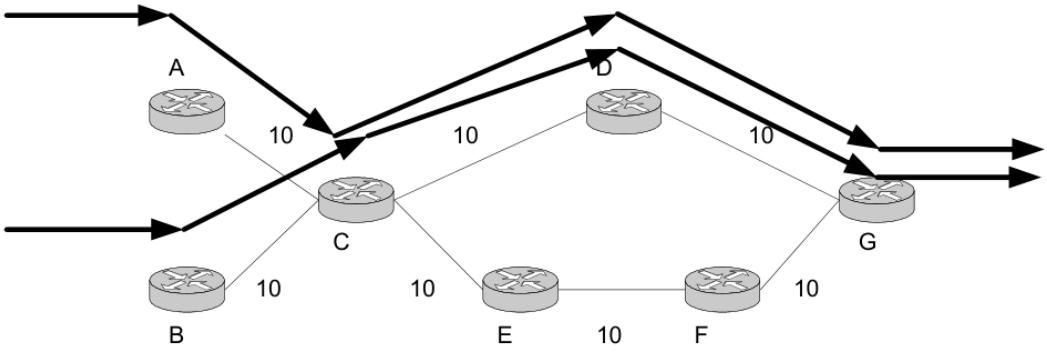


FIGURE 15 – before MPLS

This figure shows how routing worked before traffic engineering in MPLS networks. IGP protocols like OSPF or RIP used simple metrics to choose the least-cost paths, which led to traffic concentration on some links, like C $\rightarrow$ D $\rightarrow$ G, while other paths remained unused. This behavior led to underutilization of resources and congestion on the most used links. Traditional protocols did not provide load balancing or efficient use of available paths. This is where MPLS Traffic Engineering comes in, using advanced criteria like bandwidth or priorities to distribute traffic optimally and improve overall network performance. This figure therefore illustrates the need for MPLS to solve these limitations.

8.2 Why Use MPLS Traffic Engineering

1. A major gool of Internet Traffic Engineering is to facilitate efficient and rliable network operations while simultaneously optimizing network resources utilisation and trafic performance.
2. The control capabilities offered by existing Internet interior gateway protocols are not adequat for traffic Engineering .
3. Thes protocols are topology driven , so bandwidth availability and traffic characteristics decisions , consequently , congestion frequently occurs ...

8.3 What is MPLS Traffic Engineering

MPLS traffic engineering can provide several benefits to network operators, such as increasing network efficiency by using available bandwidth more efficiently and avoiding unused links. In addition, it can improve network resilience by providing fast rerouting and backup paths in case of link failure or congestion. It can also support different service levels and traffic classes by applying different QoS policies and priorities to different LSPs. In addition, it can simplify network management by reducing the complexity of routing protocols and configuration.

MPLS TE also helps in improving network reliability by allowing engineers to set up redundant tunnels in case of link failure or congestion. It also allows them to prioritize certain types of traffic, such as voice or video, over others.

By using MPLS TE, network engineers can ensure that their networks are running efficiently and reliably at all times.

MPLS Te makes forwarding descisions based on additional mateics such as :

1. Bandwidth Availability .
2. shared Risk link Groups (SRLG) .
3. Manual path selection .
4. Service Type (QOS based forwarding).
5. Affinity (link coloring).

In Other big advantage of TE :
MPLS TE also offers fast reconvergence around failures

8.4 How to Learn MPLS Traffic Engineering

MPLS traffic engineering is a complex and advanced subject that requires a solid understanding of MPLS fundamentals, routing protocols, network design, and performance analysis. To learn MPLS traffic engineering, it is important to review the basic concepts and principles of MPLS, such as label switching and label stacks. Additionally, it is worth studying the different signaling protocols and mechanisms used to establish and manage LSPs, such as RSVP-TE and LDP-TE. Additionally, it is essential to learn how to configure and troubleshoot MPLS traffic engineering on different network devices and platforms. It is also recommended to practice MPLS traffic engineering scenarios and labs using simulation tools, such as GNS3 or EVE-NG. Finally, it is important to explore the different applications and use cases of MPLS traffic engineering to gain a comprehensive understanding of the subject.

8.5 What is the role of traffic engineering

Traffic engineering refers to the set of techniques applied to control and regulate the distribution of traffic in a network. Traffic engineering therefore addresses the problem of optimizing network resources .

8.6 How MPLS Traffic Engineering Works

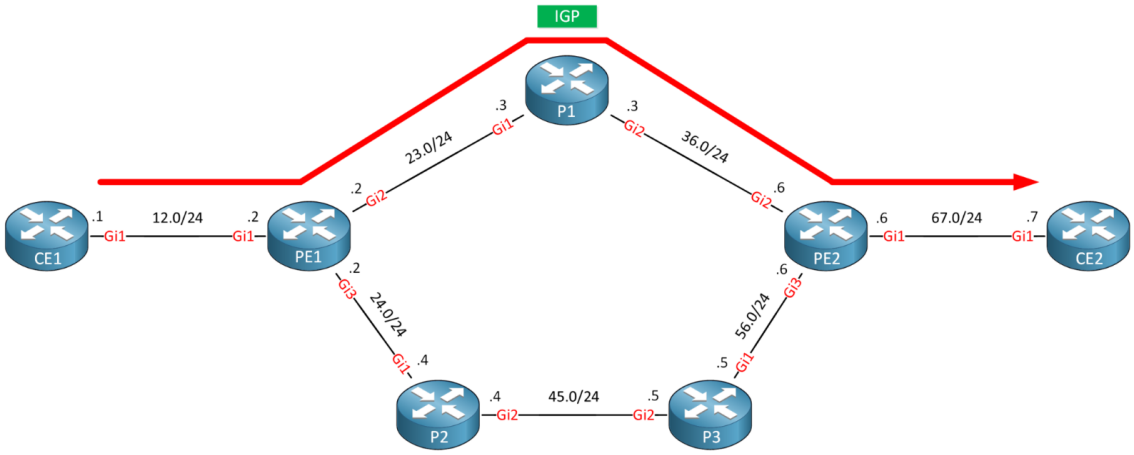


FIGURE 16 – IGP Routing

To use the network as efficiently as possible, we need to know the conditions of the network. You need to know which interfaces are utilized or underutilized. You need to know the available bandwidth in the network

With traditional IP routing and IGPs the least-cost path is determined using a metric which is a static value based on the interface is bandwidth However this approach only utilizes the path with the lowest cost disregarding alternative routes. If two paths share the same metric equal-cost load balancing can be applied, but beyond that other links remain unused leading to wasted bandwidth The only way to redirect traffic is by adjusting the metric but this solution is not always practical or feasible.

MPLS TE uses "constraint-based routing" which means that we use the shortest path that meets certain criteria. This could be a bandwidth requirement, or perhaps you want to include/exclude certain interfaces or routers in the path.

8.7 TE Tunnel basics

- TE tunel interfaces are unidirectional logical links from one router to another router
- Once the tunnel is properly configured , a label is assigned for the tunel that corresponds to the path througth the mpls network (LSP)
- The path through the network will be deterministic, and not necessarily the IGP shortest path .
- Once traffic is routed onto the tunnel the trafic flows through the tunnel based on the path .

8.8 What are the different MPLS traffic engineering techniques

Different techniques can be used to implement MPLS traffic engineering, depending on the network topology, traffic patterns, and performance objectives. Constraint-based

routing is a technique that uses a routing algorithm to find the optimal LSP for each traffic flow. This can be done either offline with a centralized controller or online with a distributed algorithm. Admission control is another technique that determines whether a new LSP request can be accepted or rejected based on the availability of network resources and the QoS requirements of the LSP. This can be done statically or dynamically. Load balancing is another technique that distributes the traffic load across multiple LSPs that have the same source and destination nodes but different paths. This can be done on a per-flow or per-packet basis. Finally, fast rerouting allows for rapid recovery from a link or node failure by switching traffic to a backup LSP that has been precomputed and reserved. This can be done locally or globally.

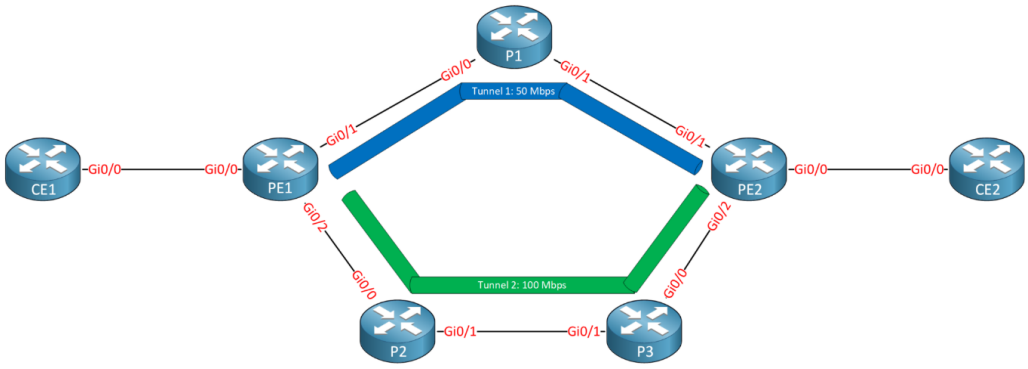


FIGURE 17 – Tunnels

To send traffic down a certain path, MPLS TE builds unidirectional tunnels from a source to the destination using Label Switched Paths (LSPs). These tunnels can have requirements such as bandwidth, affinity, or priority.

8.9 Components

1. Link information distribution .
2. Path calculation (CSPF) .
3. Path Setup (RSVP-TE) .
4. Forwarding traffic through TE tunnels .

8.10 Quality of Service (QoS) in a Network with MPLS

With MPLS, QoS is provided in a slightly different way. Packets still have their DS-fields set at the edge of the network. In addition, the experimental fields in the MPLS headers are set at the ingress LSRs. In the middle of an LSP, packets are buffered and scheduled in accordance to the experimental fields. Whether MPLS is involved or not in providing QoS is transparent to end-users. Sometimes it is desirable to use different LSPs for different classes of traffic. The effect is that the physical network is divided into multiple virtual networks, one per class. These virtual networks may have different topology and resources. The end effect is that premium traffic can use more resources than best effort traffic. Premium traffic will also have higher priority in getting the backup

resources in the case of link or router failure.

This process is largely automatic, driven by Constraint-based Routing. It is envisioned that LSPs for VPN will be built between ARs, BRs, HRs across the network in the near future. Such LSPs will be built on the top of the current LSPs that are used for Traffic Engineering. In other words, a current LSP will be treated as a link in building the LSPs for VPN. Only the endpoints of current LSPs will be involved in the signaling process of building new LSPs for VPN. LSPs are therefore stacked. Such a scheme can significantly reduce the state information in the core.

9 MPLS Fault tolerance

A network is described as fault-tolerant or survivable when it is capable of sustaining or rapidly reinstating an acceptable performance level in the event of network failures. Among the significant failures that a network may encounter are :

9.1 Types of failures :

— **Path Failure (PF)**
Refers to a defect that signals an MPLS-based recovery mechanism regarding the loss of path connectivity. A path continuity check may identify this issue. Test involving the PSL and PML. Among the various paths, some may be the most prevalent. A link probing mechanism can identify failures among neighboring LSRs. One instance of such a probing mechanism is a liveness message that is To ensure the effectiveness of either a path continuity test or a link probing mechanism, it is essential that the test message is guaranteed to traverse the same route as the working or recovery path along the segment being evaluated, exchanged periodically between peer LSRs [MPLS-PATH].

— **Failure of a link**
Due to a cut in the fibre cable. The traffic carrying cable of an office telecommunication to another is buried about three feet underground in a conduit, but because of the In construction activities, accidental cable cuts frequently occur, despite the Increased network care and maintenance efforts.

— **Node failure**
Due to equipment failure or property damage resulting from an event such as an accident fire, flood or earthquake ; consequently, all or part of the communication links connecting to the affected node may fail.

To deal with these failures, there are some requirements to think about :

- Rapid detection of faults
- Notifications mechanisms
- Recovery strategies
- Performance considerations

9.2 Failure detection/notification :

MPLS introduces a new layer in the network model that is independent of other layers used in the network. Therefore, MPLS must also have mechanisms for failure detection. Even if the lower layers have fast failure detection mechanisms, nothing can be assumed about those layers because MPLS is supposed to work with many different network types.

9.2.1 RSVP-TE Hello Extension

The Hello tension of the RSVP allows LSRs to identify when a neighboring node is unreachable. This extension consists of a HELLO message and a Hello message. A HELLO ACK object and a REQUEST object. Processing of Hello occurs between two. The failure detection interval for adjacent LSRs is configured independently, with a default setting

of 5 ms. This process involves gathering and retaining the “instance” value of an adjacent LSR. A change in this value indicates a potential issue. The neighbor is assumed to have reset, as recorded by each LSR periodically. Hello messages are generated that include the HELLO REQUEST object, utilizing the "instance" value from the LSR and the previously received "instance" value from the neighboring node. Upon receiving the message, the neighboring node conducts a comparison. transmit the "instance" value of the senders to the most recently received value from that LSR. It additionally verifies whether the sender sends back the most recent "instance" value utilized by the receiver. If any of these values are different, the neighbor is considered to have been reset. The receiver of the HELLO REQUEST then generates a HELLO ACK message. If the received values are correct, the ACK message contains the "instance" values of the sender and the receiver. Otherwise, the sender of the ACK sets the receiver's value to zero to indicate that communication has been lost. If no Hello message is received within the configured Hello time interval, the node considers that it cannot communicate with its neighbor. The default value of this hello interval number is set to 3.5 milliseconds. Therefore, the Hello protocol can detect whether the LSR is unreachable or has been restarted. Used to check whether the control plane of the LSR is working and whether the LSR is reachable. This mechanism cannot be used to check the forwarding level.

9.2.2 Graceful restart

As described earlier MPLS routers has two architectural planes, the con- trol plane and the forwarding plane. These two planes are handled by two different processors. The dedicated route processor calculates routes based from information from neighbors and then downloads these routes to the forwarding table. Forwarding can either be centralized in one processor or distributed among multiple line cards. The routing functionality is referred to as the control plane and the forwarding functionality as the forwarding plane. This separation makes it possible for a router to still forward packets correct even if the control plane is temporarily down, because the forward- ing table still contains valid routes regardless of the condition of the route processor. But when the control plane in a router goes down and restarts the control channel between the router and its adjacent router restarts. Then all LSP's traversing the router is terminated because the control channel is cut. To prevent this, a technique called graceful restart is used. In graceful restart for RSVP-TE a new object called a Restart cap object is included in the hello message. This object includes two fields called Restart time and Recovery Time. The Restart time field is set to the time it takes for a router to restart its control plane until it can start to send hello messages again. The Recovery time is set to the time a router needs to re-synchronize RSVP and MPLS forwarding state after the re-establishment of Hello syn- chronization. A value of zero indicates that the MPLS forwarding state was not preserved after the reboot. When a node detects that RSVP commu- nication with a neighbor has been lost, the node waits the amount of time indicated by the Restart time before invoking procedures for communication loss. During this period the node preserves the RSVP and MPLS state for es- tablished LSPs that traverse the restarting node. The adjacent nodes behave as if they still receive RSVP messages from the restarting node. When the node has restarted it starts to send RSVP messages again and the adjacent nodes can then resynchronize the control channel again.

9.2.3 Failure reports from other layers

Failures can also be detected by lower-layer mechanisms and reported to the MPLS layer. In this case, it is important that the recovery mechanisms are not implemented in the lower layers. In this case, recovery is performed at both the lower layers and the MPLS layer. This means that MPLS responds when an error is detected, but by the time the recovery process through MPLS is complete, the lower layers have completed their own recovery mechanisms and the connection will be working again. MPLS detects that the failed component is working again and the LSP can be redirected back to the original working path. Therefore, it is important that when recovery is implemented at the lower layers, errors that can be recovered at these layers are not reported to the MPLS layer. The error detection mechanisms at the lower layers are generally faster than those associated with MPLS. It is therefore often better to use this kind of failure detection mechanism if available.

9.3 Recovery strategies

Like the recovery mechanisms at other layers, MPLS recovery works by rerouting traffic along a new path around the point of network failure. Where this path is placed, when it is calculated, and how it is set up depends on the recovery mechanism used. If no recovery mechanism is used in an MPLS domain, recovery is performed through standard actions of the signaling protocol used to establish and maintain the LSP. This is called best effort recovery. For RSVP-TE, errors can be detected through Hello messages or through the RSVP-TE soft state mechanism. As mentioned earlier, the soft state mechanism in RSVP-TE preserves the LSP by periodically sending PATH and RESV messages. If an LSP encounters an error, the soft state cannot update the reservation, and a PathErr or ResvErr message is sent back to the input LSR for that LSP, depending on whether the PATH or RESV message failed.If an error is detected in the Hello message state, a PathErr is sent back to the input LSR. If an error is detected and reported to the input LSP, it will try to establish the LSP again. If the LSP is loosely routed, the input may try to find an output path that is separate from the link and node and try to establish the LSP on the new path. Or try to reestablish the same LSP and let the nodes near the failure point calculate a new path from the failure point to the egress. Once the path is established, traffic can flow over the new LSP. If the LSP is already strictly routed, the ingress will periodically try to reestablish the path until the path becomes available. The recovery time of this scheme is usually long and is only acceptable for best-effort type traffic. If recovery is to be performed faster, other MPLS mechanisms must be used that can try to minimize the failure notification time or the backup path calculation time.
The path recovery mechanisms in MPLS can be classified according to two criteria :

- recovery by rerouting vs. protection switching.
- local repair vs. global repair.

9.3.1 Rerouting

In the rerouting, the alternative path is calculated dynamically after a fault has been detected. After setting up the alternative LSP, the data stream is switched to the new path. Rerouting is defined as the creation of new paths or path segments on the request

to restore traffic after a fault occurs. New paths can be based on Information on faults, routing policies, predefined configurations and information about the network topology. Thus, when detecting a fault, paths or path segments are established by means of signaling to bypass the defect. For this purpose, a Replacement or backup path outside the primary path is required

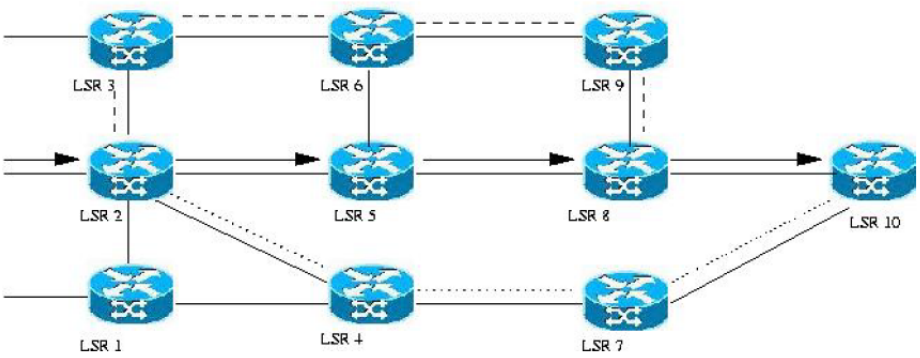


FIGURE 18 – Rerouting in MPLS networks

In rerouting there is no guarantee that a recovery path will be found. If resources needed for the recovery path are used up in the network and the recovery path has lower priority then the traffic currently using the resources, then a recovery path might not be found.

9.3.2 Protection switching

Protection switching provides fast recovery versus rerouting technique, because the alternate path is already preconfigured, so it is done immediately after the default is detected. It preestablishes an alternative path or path segment based on network policies, routing, requirements for restoring the transmission on the path and considerations administrative. When a fault is detected, the protection traffic is switched to the alternate path.In the fast restore model, the backup LSP is created and configured in advance, so the Bandwidth reservation must be done. There are two mechanisms of protection switching are : local repair (or protection link/node) and the overall repair (or path protection).

Global repair :

In global repair, protection is activated on an end-to-end basis, regardless of location of the failure, as shown in the figure below. That is, a variant or an LSP backup is preestablished from the penetration of exit routers from the path to be protected. In this case, the path alternative provides protection against all the links and nodes used in the path that is in train to route the data. When a fault is detected, the LSR must send messages from the FIS to the LSR penetration, then the recovery process will begin [26]. We notice that in the global repair, a failure signal spreads to the source (input router) before switching of traffic to the backup path, which wastes valuable time because the failure notification must cross the entire MPLS (Domain) network.

Local repair

The alternative route can be local, this means that the repair is carried out against a failure of a single link or node. In this case, FIS messages are not used as shown in *Figure*

19. This is done within a very short distance of failure, thus minimizing delays and loss total package. If a local repair is attempted to protect a set of LSPs, each intermediate LSR must have the ability to initiate a pre-established alternative PSL. indeed, it is impossible to predict where an incident can occur in an LSP. The very high cost must be paid in terms of complex calculations with extensive signaling required to establish alternative LSPs from each intermediate LSR.

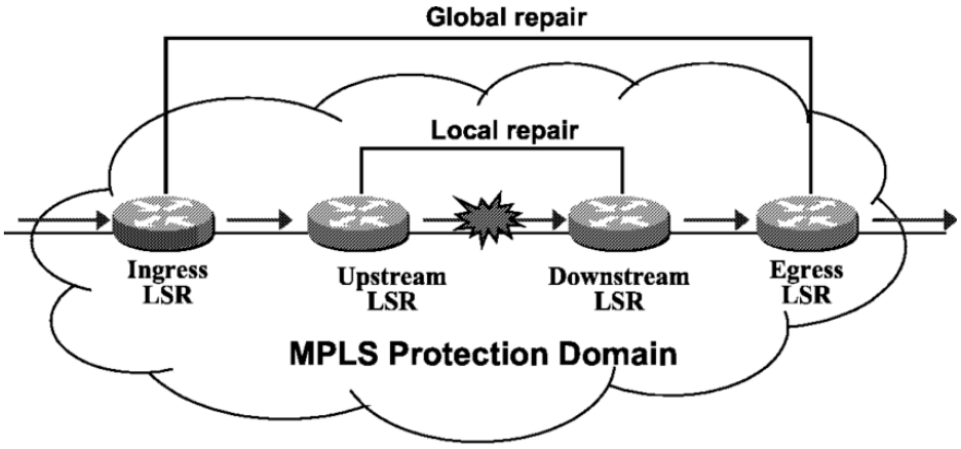


FIGURE 19 – Local vs. global repair

9.3.3 Fast Reroute FRR

When local link and node failures are detected, MPLS FRR offers a way to swiftly divert traffic from protected TE LSPs onto the backup tunnels and set up backup LSPs. MPLS FRR may reroute traffic in tens of milliseconds since the backup LSPs, also known as bypass tunnels, are set up before to the failure and offer local protection. The set of protected TE LSPs is promptly rerouted onto the proper bypass tunnel around the failed network elements when the node immediately upstream of the failure point (also referred to as the point of local repair, or PLR) detects a network element failure (either via a layer 2 mechanism or by using IGP/RSVP Hellos). An LSP is considered protected at a specific node if it possesses one or more associated backup tunnels that start at that node. In addition to a backup path, MPLS FRR offers a backup path. To put it briefly, an MPLS FRR-based local protection sheme offers a realistic, efficient, and organic defence against node or link failures. This is a comparison of MPLS FRR's bypass and diversion strategies.

	Bypass Approach	Detour Approach
Link and node failures	provides against link and node failures	Provides protection against link and node failures
LSP protection	Protects multiple LSPs by using a single bypass tunnel. offers a facility (or one-for-n) LSP protection	Requires a separate backup LSP for each primary LSP that needs primary LSP that needs to be protected, in other words, detour provides a one-for-one LSP protection
Label stacks	Employs label stack to nest multiple LSPs	Does not use label stack

Table 1 – Comparison between FRR approaches

9.4 The restoration factors in MPLS :

The most important criteria to consider when using Recovery in MPLS are :

- ◇ **Packet loss :** Restore systems may introduce packet losses during the switching or LSP backup recovery. Therefore, the rates achieved for service are So recovery systems should ensure a minimum or, if possible, no Packet loss during the restore period.
- ◇ **Packet Redirection :** Restore systems may require re-ordered packets when of switching to or from the working LSP. The package reorganization is mainly necessary At the exit of the MPLS network.
- ◇ **Resource utilization :** to provide fault tolerance in a MPLS network, particularly the Pre-planned protection, redundant bandwidth must be reserved and used efficiently.
- ◇ **The restore time :** this is the time from the beginning of failure detection, until the moment when the packages start appearing in the alternative path.

After presenting the different techniques of restoration, rerouting, and switching protection, we will present security in an MPLS network, its different techniques, and the defense strategies to be used.

10 Security measures for MPLS

With the increasing deployment of MPLS, security concerns have been raised. The basic architecture of MPLS networks does not support security mechanisms due to emerging technology MPLS to high speed packet delivery. In MPLS networks, only the two input and output routers are responsible for checking of the received IP packets header. In general, the MPLS network addresses security issues such as : confidentiality, data integrity and availability .

1. *Confidentiality :* It ensures that information is not accessible to non-commercial individuals (or entities) . In MPLS networks, it may refer to the confidentiality of LIB (Label Information Base) and the confidentiality of traffic passing through the infrastructure. MPLS to ensure that none of the labelled packages are accepted from

outside of the MPLS infrastructure, to avoid the introduction of packets by intruders that determines the value of a label. MPLS is commonly advertised as offering good MPLS VPN functionality. But Unlike VPN technology that uses IPSec, for example, MPLS VPN does not data privacy, but it allows for the use of encrypted protocols such as HTTPS and IPSec. An interesting feature of MPLS is that it allows network operators to hide details on their MPLS core. Even if a packet is passing through the network, only its labels are subject to manipulations.

2. *Data integrity :* Data integrity is the fact that information has not been modified or destroyed in an unauthorized manner. MPLS relies on trust in the packet transfer decisions made by LIB. Updates to LDP information should only be accepted from reliable sources. This can be done through two mechanisms : First, LDP updates should only be accepted from the interface where another LSR is known. In other words, LDP updates should not be accepted by clients outside of the MPLS kernel. Then, if the core routers are not trusted or are supposed to be vulnerable to attack, authentication mechanisms must be in place to protect the LDP.
3. *Availability :*It is the property of a system or system resource to be accessible or usable by the request of an authorized entity. Thus, network operators must maintain devices within basic infrastructure.

10.1 Security threats in MPLS :

Now, we will discuss the various network security threats that can endanger the MPLS network.
Several types of attacks exist :

— Attacks on the control plane.

— Attacks on the data plane.

— Attacks of the MPLS core interiors.

10.1.1 Attacks on the control plane :

This category includes attacks on control structures operated by the service provider of the MPLS kernel. These attacks include :

Unauthorized Element Creation of LSP :
The unauthorized item may be a local CE (CustumerEdge) or a router in another domain. Unauthorized elements can generate signaling of MPLS tags, which may result in the plan control, in the state of transmission in case of success, as well as an unnecessary reservation of the band network. This can also result in theft of service or even risk the entire network.

Interception of LSP message :
This threat can be accomplished by network traffic monitoring after a physical intrusion. Thus without physical intrusion, it could be accomplished with unauthorized modification of software. Also, many technologies such as terrestrial microwave, satellite, or fiber optic could be intercepted without physical intrusion. If successful, it could provide information leading to the label.

Attacks against LDP :
LDP is the control protocol used to set up MPLS tunnels without TE (Traffic Engineering). There are two important types of attack against the LDP : An unauthorized

network element can establish an LDP session by sending LDP Hello messages and LDP Init, leading to the potential implementation of an LSP, as well as the LDP. An attack can launch a DoS (Denial of Service) attack in the form of a message storm LDP Hello or LDP TCP SYN, leading to the use of target router resources.

Unauthorized Access to Service Provider Equipment :
This involves gaining unauthorized access to a service provider's infrastructure, enabling actions such as reconfiguring equipment or extracting network-related information like statistics and topology.

User Traffic Cross-Linking :
This situation occurs when the separation between user traffic is compromised. It can manifest in various ways, including :
- A site being incorrectly connected to a different VPN.
- Duplication of traffic forwarded to an unauthorized user.
- Merging of multiple VPNs that should remain distinct.
- Creation of a point-to-point VPN linking two inappropriate endpoints.
- Delivery of packets or frames outside their intended VPN.

Such cross-linking may stem from the service provider's actions, equipment malfunctions, or intentional attacks. Breaches can be physical, like poorly connected PE-CE links, or logical, such as improper device configurations.

10.1.2 Data Attacks :

This section covers attacks aimed at either the service provider or the end user regarding data. From the perspective of users within an MPLS network, this could include disruptions in the traffic control plan, such as routing protocols transmitting data from User A's site to User B's across IP or non-IP connections, potentially involving specific types of VPNs.
These data attacks can include :

— Unauthorized observation of traffic data
— Change in traffic data
— Unauthorized insertion of data traffic
— Unauthorized deletion of data
— Unauthorized traffic analysis
— denial-of-service attacks

10.1.3 Security Defense means

Here we will present some of the security weapons used here Some security threats can be addressed. These techniques described here include authentication, encryption, access control, filtering, firewall, isolation, aggregation and others

— **Authentication :** To avoid security problems arising from certain DoS attacks or from malicious or from accidental misconfiguration, it is essential that devices in the MPLS do not accept that connections or check messages from valid sources. Authentication refers to Methods to ensure that message sources are properly identified

by the MPLS devices with which they communicate. There are several strategies in MPLS to ensure authentication, we quote :

- **Authentication management system :** The authentication management system includes authentication of a PE to a manager of network that is centrally managed or to a directory server. It also includes a authentication of CE to configuration server, when a configuration server system is used. It should be bidirectional.
- **Point-to-point authentication :** Point-to-point authentication includes peer authentication in the network of Control (for example, LDP, BGP, etc.) and other peers. Authentication should be bidirectional, Configuration server PE or CE to be sure it is in communication with the good server.
- **Cryptographic techniques for authenticating identity :** Cryptographic techniques offer several mechanisms for authentication, device identity or individuals. These include the use of shared secret keys, unique generated keys by accessory or software devices, user name and password pairs and a range of public-private keys. Another approach is to use a certification authority system hierarchical and provide digital certificates.

— **Cryptographic techniques :** There are several cryptographic techniques used in MPLS, we quote :

- **IPSec in MPLS :** IPSec is the security protocol of choice for IP layer protection. It provides robust security for IP traffic between device pairs. Non-IP traffic, such as IS-IS routing must be converted to IP (for example, by encapsulation) in order to use IPSec. When the MPLS encapsulates the IP traffic, IPSec covers the encryption of the client's IP layer.

  In the MPLS model, IPsec can be used to protect IP traffic between SPEs, between a PE and a CE or CE to CE. Similarly, it can use a variety of integrity or confidentiality algorithms with various key lengths, such as AES encryption. There is a compromise between the key length, the the computing load, and the security level of the encryption.
- **Encryptions for device configuration and management :** For configuration and management of MPLS devices, encryption and authentication of the connection management at a level comparable to that provided by IPsec is desirable. Several methods of traffic transport and MPLS device management provide authentication, data integrity and Confidentiality, we quote : Secure SHell (SSH) : it provides protection for Telnet to allow remote configuration. Transport Layer Security (TLS) : it is based on HTTPS, it is widely used to secure the communication and can thus provide support for most on XML and management approaches SOAP based devices ; IPSec : provides security services, including data integrity and confidentiality of the network layer

— **Access control techniques :** Access control techniques include packet-by-packet provides access through filters and firewall on IPv4 and IPv6 packets, as well as by means of session admission for the control, signaling or management protocol. In this section we distinguish two techniques, filtering and firewall.

- **Filtering :** It is relatively common for routers to filter packets. These routers can look in the packet header. Packets that meet the criteria associated with a particular filter can be either rejected or receive special treatment. Today, not

only routers but also Hosts have filters and each instance of IPSec also has a filter.

— **Firewall :** Firewalls provide a mechanism to control traffic passing between different trusted zones in MPLS or between a large area and an unsecured area. Firewalls typically provide much more functionality than filters, as they may be able to apply, analyze and detail the flow logic functions, not just individual packets. They can offer a variety of complex services, such as protection against DoS attacks, virus detection, etc. As with other access control techniques, the value of firewall depends on a clear understanding of Topology of the MPLS core network, user networks and threat model.

11 Comparison between MPLS and SD-WAN

11.1 What is SD-WAN ?

A software-based method of WAN management is called software-defined wide-area networking, or SD-WAN. Similar to MPLS, businesses use it to connect their networks across several locations, including distant branches, data centers, cloud instances, and the devices of remote workers. An advancement in connectivity over conventional MPLS technology is SD-WAN. Customers can use a range of transport options, including MPLS circuits, direct internet broadband, and LTE/5G, and prioritise important business traffic. It makes the WAN independent of transport tiers and streamlines its administration.
More organizations are upgrading to SD-WAN because of the rapid rise of remote work and cloud adoption. SD-WAN provides edge users with more secure, reliable network connections when they access enterprise applications and data anytime, anywhere, while delivering a predictable user experience. SD-WAN enables enterprises to prioritize data based on its type and choose the shortest, fastest, most reliable, and cost-effective path.

11.2 What are some benefits of SD-WAN compared to MPLS ?

- **SD-WANs do not rely on specialized hardware**, MPLS requires configuring specialized routers to forward packets correctly. SD-WANs can run using any networking hardware.
- **SD-WANs have no inherent bandwidth limits.** Because MPLS connections are more or less set in stone (unless they are reconfigured), there is a hard limit over how much capacity can be provisioned over an MPLS connection at once. SD-WAN connections can add capacity as required by combining multiple connections and leveraging the fastest connectivity available.
- **SD-WANs are service provider-agnostic.** MPLS requires organizations to use the same carrier at all WAN-connected sites because MPLS connections have to be configured in physical routers in the adjacent network.
- **SD-WAN connections run over the regular Internet ;** any ISP can support an SD-WAN connection.
- **SD-WAN routing is more flexible.** SD-WAN can take advantage of multiple connectivity options including broadband Internet connections, private lines, and 5G. It can direct traffic and failover between all available connectivity options.

MPLS services typically require dedicated private line connections from the service provider.

- **SD-WANs integrate more easily with the cloud.** Connecting to the cloud via MPLS is a specialized service offered by some MPLS service providers for some cloud providers. With MPLS, connecting with the cloud requires constructing a direct route to that cloud provider's infrastructure.

11.3 What are some SD-WAN drawbacks compared to MPLS ?

- **MPLS offers more granular control over where packets go**
  Typically, data packets on the Internet take different routes depending on how routers on the path are forwarding packets at that time — but MPLS routes are only updated manually. MPLS packets never deviate from the defined route since they always are forwarded between the same routers and networks. Depending on the routing method used, network traffic on an SD-WAN likely will not always take the same route, and some packets may be lost in transit, as is the case with most Internet traffic.
- **MPLS is sometimes more reliable** MPLS traffic is usually given a higher priority over service provider backbone networks relative to Internet traffic. MPLS services often include quality-of-service (QoS) guarantees. SD-WAN leverages best effort Internet services and may experience occasional packet loss. However, most SD-WAN services compensate for this by intelligently steering traffic away from lossy connections.
- **MPLS is better for real-time applications.** MPLS often includes multiple classes of service which are preserved throughout the WAN. This makes it easier to ensure real-time applications have a better experience across the WAN.

11.4 Does SD-WAN replace MPLS ?

For many businesses, MPLS remains a relevant and efficient solution for network connectivity.

Organizations in certain industries, such as defense or healthcare, may need to maintain MPLS infrastructure for compliance and security reasons and not move their vital data and apps to the cloud. They may want to use MPLS to ensure a high level of reliability in network connectivity to support certain business-critical applications.

While the move to SD-WAN is accelerating rapidly, most companies are in a hybrid state right now. In other words, they are using both MPLS and SD-WAN solutions where they make the most sense for the business and its users.

Some organizations also hesitate to move too quickly to new technology. Transitioning away from legacy, technologies like MPLS takes time and effort. So, some businesses that take a hybrid approach do so because it lets them expand their use of SD-WAN at a comfortable pace while increasing agility, flexibility, and security in their network.

12 Conclusion :

The Multi-Protocol Label Switching (MPLS) protocol has revolutionized traffic network management by introducing a novel design that prioritizes speed, flexibility, and security. MPLS provides durable and efficient solutions that are well suited to contemporary network environments by using étiquettes to optimize routing. This technology makes it possible to implement sophisticated traffic engineering mechanisms, effectively manage private virtual networks (VPNs), and ensure high resilience against panes. These features make MPLS an essential solution to meet the growing demands of businesses in terms of network performance, reliability, and quality of service (QoS). As a result, MPLS is essential to the development of contemporary communication infrastructures, supporting their ability to meet present and future challenges.

13 MPLS Configuration

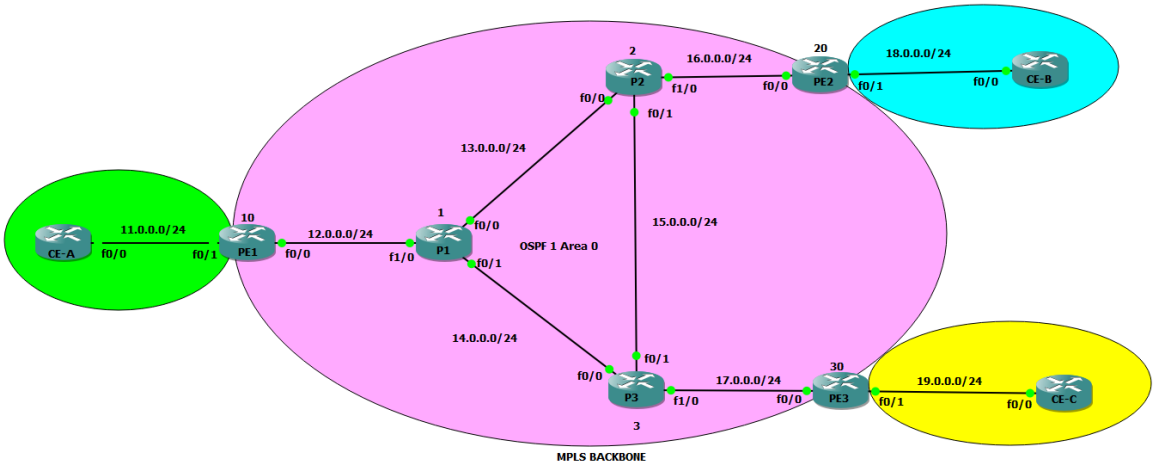


FIGURE 20 – MPLS Backbone

13.1 Basic configuration

Avant de commencer la configuration de ce laboratoire, j'aimerais expliquer une petite différence entre les routeurs P (Provider) et PE (Provider Edge). Tout simplement, le routeur P (Provider) ne traite pas directement les paquets, il agit uniquement comme un pont pour tracer un chemin entre la source et la destination. En revanche, le routeur PE (Provider Edge) est responsable d'ajouter, de modifier ou de retirer les étiquettes (labels).

Le routeur P (Provider) sert simplement à faire transiter le trafic à travers le réseau. Il ne peut ni ajouter ni supprimer des étiquettes (labels), ni effectuer d'autres modifications sur celles-ci.

Le routeur PE (Provider Edge) assure la communication entre le réseau MPLS (ou le backbone MPLS) et le réseau du client. C'est ce qu'on appelle communément un Customer Edge (CE).

Donc La communication entre le routeur CE (Customer Edge) et le backbone MPLS passe toujours par les PE (Provider Edge). Ainsi, le routeur PE agit comme un intermédiaire entre le client (Customer) et le backbone MPLS.

LDP (Label Distribution Protocol) : Le protocole responsable de la distribution des étiquettes .

La communication entre les routeurs P se fait toujours via des protocoles de routage interne (IGP). En revanche, la communication entre les PE s'effectue toujours via BGP (Border Gateway Protocol), en s'appuyant sur des tunnels.

13.1.1 Hostname and Addressing (Backbone interfaces)

On va commencer par la configuration interne je parle sur la configuration de Back-Bone ou bien MPLS Network. PE1 and PE2 and PE3 and P1 and P2 and P3 .

La configuration de PE1 et PE2

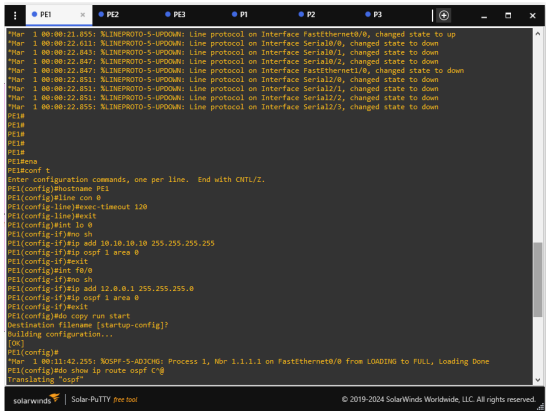


FIGURE 21 – PE1 Config



FIGURE 22 – PE2 Config

La configuration de PE3 et P1



FIGURE 23 – PE3 Config



FIGURE 24 – P1 Config

La configuration de P2



FIGURE 25 – P2 Config



FIGURE 26 – P2 Config

La configuration de P3



FIGURE 27 – P3 Config



FIGURE 28 – P3 Config

13.1.2 Routing (Backbone zone) : OSPF



FIGURE 29 – Backbone zone

13.2 Test

le commande : do show ip route ospf



FIGURE 30 – Routing table

La table de routage est correcte. Après cette configuration, il est important de la vérifier pour s'assurer que tout fonctionne correctement.

13.2.1 Ping frome PE1 to PE2 and PE3

pour le test on peut utiliser les deux commandes suivant :

1. do ping 20.20.20.20
2. do ping 30.30.30.30



FIGURE 31 – Ping

Ping vers les addresses 20.20.20.20 c'est PE2 et aussi vers 30.30.30.30 c'est PE3 , juste pour simplifier le diagnostic et si il y a des erreurs on doit les corriger .

ospf 1 area 0 : la communication entre les retours dans le bacbone , mais la communication entre les CE via BGP , BGP entre PE1 , PE2 et PE3 .

13.3 Routing table

la commande : **do show ip route ospf**

13.4 LDP configuration

Pour la configuration de LDP (label Distributed Protocol) on utilise les commandes suivantes :



FIGURE 32 – Ping

1. router ospf 1
2. mpls ldp autoconfig
3. exit

il faut tapez ces commandes sur PE1 , PE2 , PE3 , P1 , P2 et P3

PE1 and PE2



FIGURE 33 – PE1



FIGURE 34 – PE2

PE3 nd P1



FIGURE 35 – PE3



FIGURE 36 – P1

P2 and P3 :



FIGURE 37 – P2



FIGURE 38 – P2

13.4.1 Test

pour tester on utiliser la commande :

–do show mpls interface



FIGURE 39 – MPLS interface

Les interfaces f0/0, f0/1 et f1/0 sont configurées avec les adresses IP, en se basant sur LDP. Jusqu'à présent, il n'y a pas de tunnel BGP configuré, et la configuration statique n'est pas encore effectuée. Cependant, la configuration opérationnelle est déjà activée.

–do show mpls ldp neighbor

Pour connaître la version de votre retour, l'identifiant est composé de 2 ou 3 parties, incluant les adresses IP et toutes les informations nécessaires, telles que le nombre de paquets envoyés, reçus, ainsi que les messages envoyés et reçu .



FIGURE 40 – MPLS ldp

13.5 MP-BGP Configuration

La configuration de MP-BGP est nécessaire pour créer des tunnels entre les routeurs PE, car il est essentiel d'échanger les adresses IP utilisées au niveau du backbone avec les clients. Par conséquent, la solution consiste à configurer MP-BGP.

Dans cette configuration de MP-BGP j'ai utilisé les commandes suivant (voir la figure 29) :

1. neighbor 20.20.20.20 remote-as 1
2. neighbor 30.30.30.30 remote-as 1
3. neighbor 20.20.20.20 update-source lo 0 *(cette commande pour envoyer le mise à jour)*
4. neighbor 30.30.30.30 update-source lo 0
5. no auto-summary *(pour désactiver les résumer des adresse)*
6. address-family vpnv4 unicast *pour lancer VPN*
7. neighbor 20.20.20.20 activate
8. neighbor 30.30.30.30 activate
9. exit
10. exit

Il faut effectuer la même configuration pour PE2 et PE3.

Remarque : Lorsque l'on utilise des adresses IP, les routes BGP offrent plusieurs choix : si une interface physique tombe en panne, le trafic peut être redirigé vers une autre route. En revanche, si l'on utilise un adressage basé uniquement sur une interface physique, en cas de panne, il n'y aura pas d'alternative pour rediriger le trafic.

13.5.1 Test

Pour le test de MP-BGP on utilise la commande : **do show bgp vpnv4 unicast all summary**

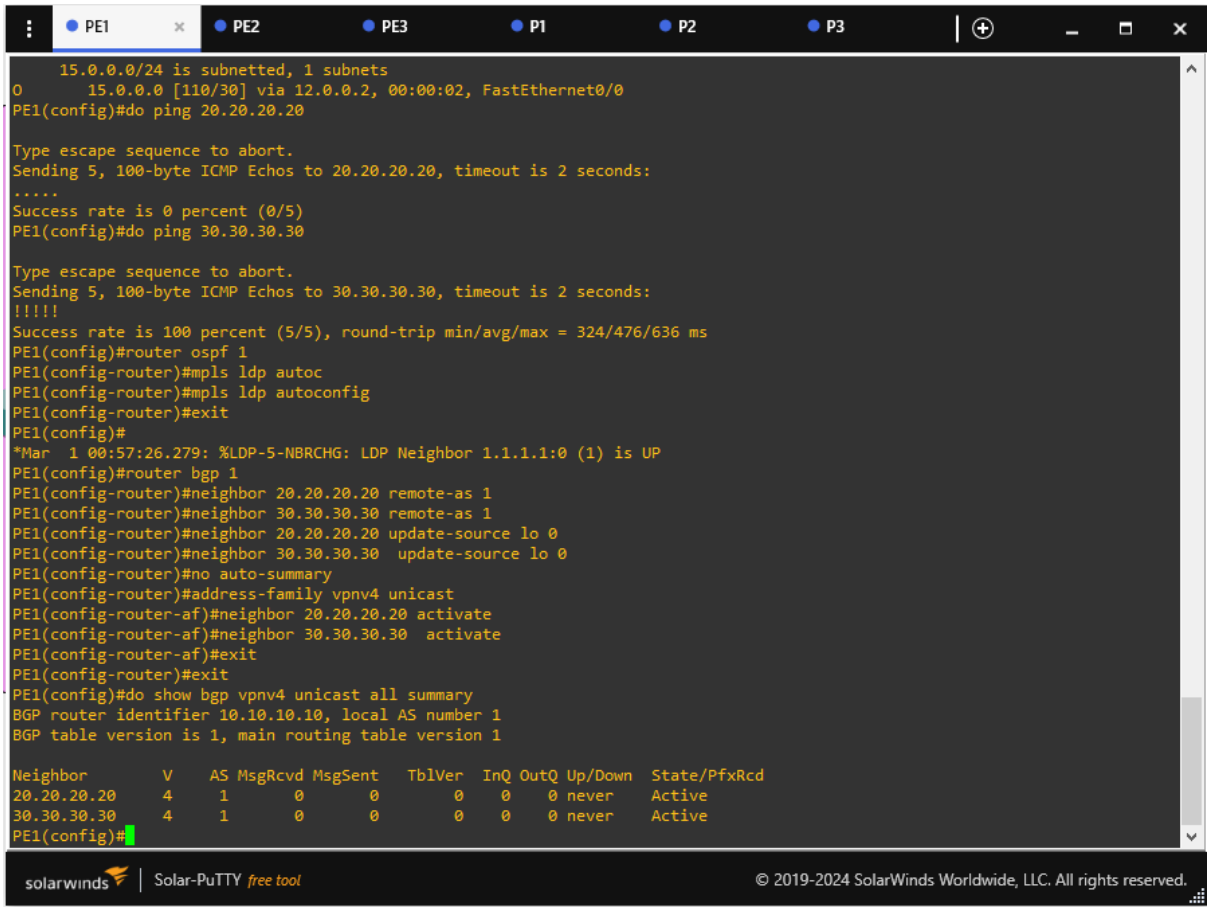


FIGURE 41 – configuration de MP-BGP et show bgp

D'après les résultats de cette commande, on remarque que tout est correct. Si une table est vide, cela indique un problème dans la configuration. Cependant, dans ce cas, tout est parfait .

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